

Exploiting the Intrinsic Neighborhood Semantic Structure for Domain Adaptation in EEG-Based Emotion Recognition

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Abstract—Due to the inherent non-stationarity and individual differences present in electroencephalogram (EEG) signals, developing a generalizable model that performs well on new subjects is challenging in EEG-based emotion recognition. Most existing domain adaptation (DA) methods typically mitigate these discrepancies by aligning the marginal distributions of domain feature representations. However, when there is a significant difference in the class-conditional distribution between domain features and labels, the domain-invariant features learned by aligning marginal distributions may have limited discriminative ability for unlabeled target instances or even prove counterproductive. To address this issue, we propose a Neighborhood Semantic Aware Learning-based Dynamic Graph Attention Convolution (NSAL-DGAT) approach that learns target semantic information by considering the inter-domain semantic topological structure, thereby improving classifier adaptation for target instances. Specifically, the proposed

NSAL framework is designed to capitalize on the insight that after domain feature alignment, some target samples and their neighboring source samples exhibit similar semantics. By leveraging the neighborhood topological structure, we extract and incorporate semantic target features to train a more transferable classifier. Besides, we implement an entropy weighting mechanism to emphasize representative target semantic information, encouraging target instances to prioritize high-confidence individuals within the source neighborhood. We have conducted extensive experiments on the public SEED dataset and our collected the Hearing-Impaired EEG Dataset (HIED). The experimental results underscore the efficacy of our proposed NSAL-DGAT approach, showcasing state-of-the-art accuracy in subject-dependent as well as subject-independent scenarios.

Index Terms—Domain adaptation (DA), emotion recognition, brain-computer interface, electroencephalogram (EEG).

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I. INTRODUCTION

EMOTION is a complex psychological and physiological state affected by an individual's subjective environment, life experience, and psychological state. It not only affects one's internal psychological state but also manifests in external behavioral expressions [1]. Understanding and studying emotions is crucial. With the development of science, psychology, and cognitive neuroscience, emotion recognition has gradually become a widely researched and important topic. Historically, researchers have mainly relied on external behavioral signals for emotion recognition, such as speech [2], [3], text [4], [5], facial expressions [6], [7], and gestures [8], [9]. However, the affective states are easily camouflaged or concealed within these non-physiological signals, making it difficult to accurately assess the true affective states. In recent years, the electroencephalogram (EEG)-based emotion recognition has garnered significant attention from researchers [10], [11]. The EEG signals contain an individual's spontaneously generated central and neural activity, which provides a more direct and precise reflection of the individual's underlying emotional state. Moreover, by analyzing and interpreting the temporal-spatial features of EEG signals, researchers can gain deeper insights into the relationship between emotion and neural activity.

In the past few decades, EEG-based emotion recognition has made significant progress and achieved remarkable results [12], [13]. Numerous graph convolutional network (GCN) methods have recently been adopted to exploit the oscillatory patterns of

基于脑电图的情绪识别中的领域自适应利用内在邻域语义结构

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摘要—由于脑电图（EEG）信号中存在固有的非平稳性和个体差异，在基于 EEG 的情绪识别中，开发一个在新受试者上表现良好的泛化模型具有挑战性。大多数现有的领域自适应（DA）方法通常通过对领域特征表示的边缘分布进行对齐来缓解这些差异。然而，当领域特征和标签之间的类条件分布存在显著差异时，通过对齐边缘分布学习到的领域不变特征可能对未标记的目标实例的判别能力有限，甚至可能适得其反。为了解决这个问题，我们提出了一种基于邻域语义感知学习的动态图注意力卷积（NSAL-DGAT）方法，该方法通过考虑领域间的语义拓扑结构来学习目标语义信息，从而提高对目标实例的分类器自适应能力。具体而言，所提出的方法

NSAL 框架旨在利用这样的见解：在域特征对齐后，一些目标样本及其邻近的源样本表现出相似的语义。通过利用邻域拓扑结构，我们提取并整合语义目标特征，以训练一个更具可迁移性的分类器。此外，我们实现了一种熵加权机制，以强调具有代表性的目标语义信息，鼓励目标实例优先考虑源邻域中的高置信度个体。我们在公开的 SEED 数据集和我们收集的听力障碍脑电数据集 (HIED) 上进行了广泛的实验。实验结果强调了我们所提出的 NSAL-DGAT 方法的有效性，展示了在受试者相关和受试者无关场景中的最先进准确率。

索引词—领域自适应 (DA)，情感识别，脑机接口，脑电图 (EEG)。

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I. I

情绪是一种复杂的心理和生理状态，受个体主观环境、生活经历和心理状态的影响。它不仅影响个体的内部心理状态，还通过外部行为表达表现出来[1]。理解和研究情绪至关重要。随着科学、心理学和认知神经科学的发展，情绪识别逐渐成为广泛研究和重要课题。历史上，研究人员主要依赖外部行为信号进行情绪识别，如语音[2],[3]、文本[4],[5]、面部表情[6],[7]和手势[8],[9]。然而，情感状态在这些非生理信号中容易被掩盖或隐藏，难以准确评估真实情感状态。近年来，基于脑电图（EEG）的情绪识别引起了研究人员的广泛关注[10],[11]。脑电图信号包含个体自发产生的中枢和神经活动，能够更直接、更精确地反映个体的潜在情绪状态。此外，通过分析和解释脑电图信号的时空间特征，研究人员可以更深入地了解情绪与神经活动之间的关系。在过去的几十年里，基于脑电图的情绪识别取得了显著进展并取得了卓越成果[12], [13]。近年来，许多图卷积网络（GCN）方法被采用来利用脑电信号的振荡模式

源代码可在 <https://github.com/YYingDL/NSAL-DGAT> 获取。数字对象标识符 10.1109/TAFFC.2025.3564272

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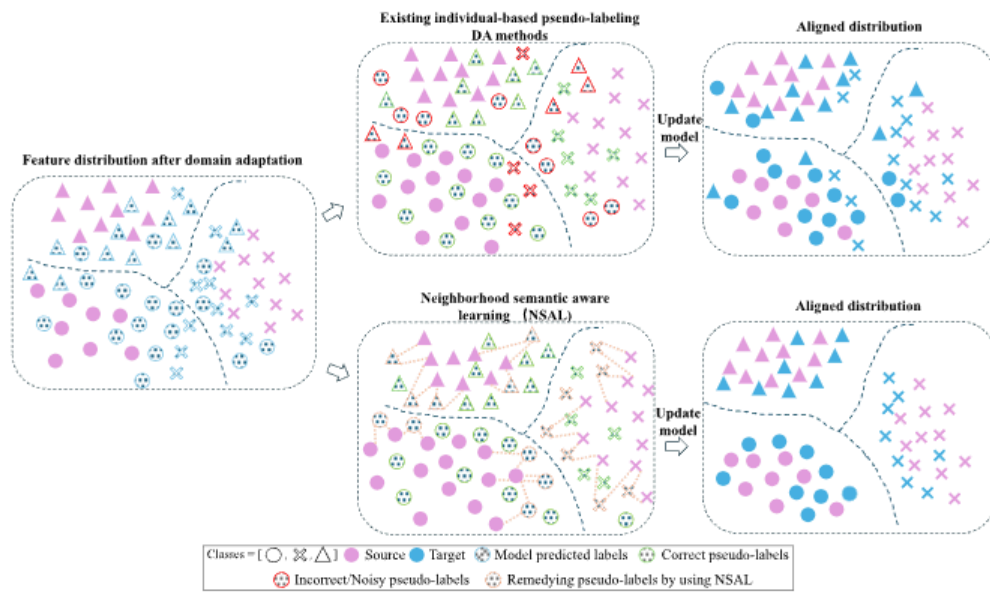


Fig. 1. Most related DA works use individual predictions to obtain pseudo labels, often overlooking the semantic topological structure. In contrast, NSAL focuses on capturing the intra-class semantic similarity between source and target domain samples to remedy pseudo labels, thereby obtaining more discriminative domain-invariant features.

neural rhythms and functional connectivity across brain regions to decode emotional states [14], [15]. However, a major problem with these studies is that most GCN-based methods focus on global graph representations while ignoring class substructures - fine-grained topological changes associated with specific emotional states (e.g., anger versus happiness). This limitation hinders learning discriminative and generalizable features, especially in subject-independent scenarios.

On the other hand, non-stationarity and individual differences pose challenges for building a classifier with strong generalization and discrimination capabilities [16]. Domain adaptation (DA) helps mitigate these discrepancies by learning domain-invariant features that enable knowledge transfer from source domains to target domains. While most existing DA methods focus on aligning global feature distributions (e.g., via adversarial training [17], [18] or distribution matching [18], [19]), they often neglect semantic structural relationships between domains. Feature alignment without explicit class-wise constraints results in fragmented intra-class clusters and overlapping inter-class boundaries across domains. Recently, a few works [20], [21] have considered class-wise semantic information to learn more discriminative features. However, as shown in Fig. 1(a), these methods merely rely on individual, threshold-based pseudo-labeling for obtaining semantic targets, overlooking the intrinsic neighborhood structure between source and target samples in the feature space after domain feature alignment, easily leading to noisy semantic information and negative transfer.

To tackle these critical issues, we propose a neighborhood semantic aware learning-based dynamic graph attention convolution (NSAL-DGAT) approach, which learns semantic targets from the neighborhood topological structure to enhance the classifier's adaptability. As shown in Fig. 1, the proposed NSAL takes inspiration from the observation that, after domain feature alignment, samples from the same class in both the source and target domains are expected to form clusters in the embedding space [20], [22]. Thus, we assume that the target sample shares semantics similar to its nearest neighbors, allowing target semantics to be learned from adjacent source

samples. Furthermore, to obtain representative target semantics, we introduce an entropy-weighted strategy that encourages target instances toward high-confidence individuals within the semantic topological structure. Finally, we incorporated semantic target features as a regularization method to train a more transferable classifier. We conducted extensive experiments on the SEED dataset and our hearing-impaired EEG dataset (HIED). These experimental results demonstrate that the proposed NSAL-DGAT method significantly improves classification accuracy in both subject-dependent and subject-independent scenarios for individuals with typical hearing and individuals with hearing loss.

Overall, the main contributions of this paper can be summarized as follows:

- We propose a Neighborhood Semantic-Aware Learning (NSAL) approach that leverages the intrinsic neighborhood semantic topological structure between source and target samples after domain feature alignment to learn the semantic information of target instances effectively.
- To acquire representative target semantic information, we use the entropy weighting strategy to encourage target instances toward high-confidence individuals in neighborhood source samples.
- Experimental results on two datasets demonstrate that the proposed NSAL-DGAT significantly improves classification accuracy in EEG-based subject-dependent and subject-independent emotion recognition.

In the following sections, this paper will be organized according to the following structure. First, Section II provides a concise introduction to the related works. Next, Section III presents the proposed NSAL-DGAT method. Section IV describes extensive experimental results. In Section V, we discuss the interpretable analysis. Lastly, Section VI provides conclusions and future work.

II. RELATED WORK

A. EEG-Based Emotion Recognition

EEG-based emotion recognition has significantly progressed and achieved encouraging results in recent years. A traditional emotion recognition method mainly comprises two parts: feature extraction and classifier design. In the feature extraction stage, researchers typically manually extract EEG features from the time domain, frequency domain, or time-frequency domain [23], [24]. For feature classification, various traditional and deep learning methods classify different emotional states. For instance, Li et al. [25] proposed a graph-regularized sparse linear regression (GRSLR) method, which extends the transformation matrix of linear regression by incorporating graph regularization and sparse regularization. Zheng et al. [26] introduced the group sparse canonical correlation analysis (GSCCA) to explore the correlation between the EEG features and emotion labels. Tao et al. [27] developed an attention-based convolutional recurrent neural network (ACRNN) that extracts crucial spatial and temporal information from EEG signals, facilitating the learning of discriminative emotion features. Li et al. [28] proposed a regional-to-global hierarchical spatial-temporal neural network

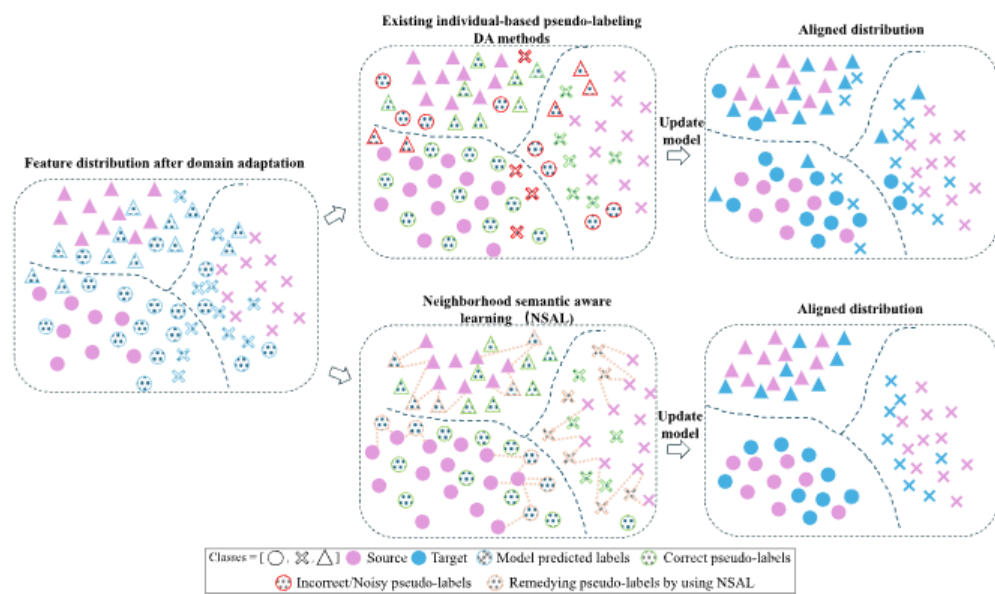


图 1. 大多数相关领域自适应 (DA) 工作使用单个预测来获取伪标签，往往忽略了语义拓扑结构。相比之下，NSAL 专注于捕获源域和目标域样本之间的类内语义相似性来修正伪标签，从而获得更具判别力的领域不变特征。

神经节律和跨脑区的功能连接来解码情绪状态[14], [15]。然而，这些研究的一个主要问题是大多数基于图卷积网络 (GCN) 的方法专注于全局图表示，而忽略了类子结构——与特定情绪状态相关的细粒度拓扑变化（例如，愤怒与快乐）。这一局限性阻碍了学习具有判别力和泛化能力的特征，特别是在受试者独立场景中。

另一方面，非平稳性和个体差异对构建具有强泛化能力和区分能力的分类器提出了挑战[16]。领域自适应 (DA) 通过学习领域不变特征，帮助缓解这些差异，实现知识从源域到目标域的迁移。虽然大多数现有的 DA 方法专注于对齐全局特征分布（例如通过对抗训练[17][18]或分布匹配[18][19]），但它们往往忽略了领域间的语义结构关系。没有显式类别约束的特征对齐会导致类内聚类碎片化以及跨领域的类间边界重叠。最近，少数工作[20][21]考虑了类别级的语义信息来学习更具区分度的特征。然而，如图 1(a)所示，这些方法仅仅依赖于个体、基于阈值的伪标签来获取语义目标，忽略了领域特征对齐后源样本和目标样本在特征空间中的内在邻域结构，容易导致噪声语义信息和负迁移。

为解决这些关键问题，我们提出了一种基于邻域语义感知学习的动态图注意力卷积 (NSAL-DGAT) 方法，该方法从邻域拓扑结构中学习语义目标以增强分类器的适应性。如图 1 所示，所提出的 NSAL 方法受到以下观察的启发：在域特征对齐后，源域和目标域中属于同一类别的样本在嵌入空间中预期会形成簇[20], [22]。因此，我们假设目标样本与其最近邻样本具有相似的语义。

允许从相邻的源样本中学习目标语义。此外，为了获得具有代表性的目标语义，我们引入了一种熵加权策略，该策略鼓励目标实例向语义拓扑结构中的高置信度个体靠拢。最后，我们将语义目标特征作为正则化方法来训练一个更可迁移的分类器。我们在 SEED 数据集和我们的听力障碍 EEG 数据集 (HIED) 上进行了广泛的实验。这些实验结果表明，所提出的 NSAL-DGAT 方法在具有正常听力的人和听力障碍的人的受试者相关和受试者无关场景中，都显著提高了分类准确率。

总的来说，本文的主要贡献可以总结如下：我们提出了一种邻域语义感知学习

一种利用领域特征对齐后源样本和目标样本之间固有的邻域语义拓扑结构，以有效学习目标实例语义信息的(NSAL)方法。为了获取具有代表性的目标语义信息，我们采用熵加权策略，鼓励目标实例向邻域源样本中的高置信度个体靠拢。在两个数据集上的实验结果表明，所提出的 NSAL-DGAT 显著提高了基于脑电图的情绪识别中的受试者相关和受试者无关的分类准确率。

在接下来的部分，本文将按照以下结构组织。首先，第二节简要介绍相关工作。接下来，第三节介绍所提出的 NSAL-DGAT 方法。第四节描述了广泛的实验结果。在第五节，我们讨论可解释性分析。最后，第六节提供结论和未来工作。

II. RW

A. 基于脑电图的情绪识别

基于脑电图的情绪识别近年来取得了显著进展，并取得了令人鼓舞的结果。传统的情绪识别方法主要包含两个部分：特征提取和分类器设计。在特征提取阶段，研究人员通常从时域、频域或时频域手动提取脑电图特征[23][24]。对于特征分类，各种传统和深度学习方法对不同情绪状态进行分类。例如，Li 等人[25]提出了一种图正则化稀疏线性回归 (GRSLR) 方法，该方法通过结合图正则化和稀疏正则化扩展了线性回归的转换矩阵。Zheng 等人[26]引入了组稀疏典型相关分析 (GSCCA) 来探索脑电图特征与情绪标签之间的相关性。Tao 等人[27]开发了一种基于注意力的卷积循环神经网络 (ACRNN)，该网络从脑电信号中提取关键的空间和时间信息，有助于学习具有区分性的情绪特征。Li 等人[28]提出了一种区域到全局的层次化时空神经网络

(R2G-STNN) method that enables hierarchical feature learning from local to global regions to learn discriminative spatiotemporal features. Despite their remarkable performance, these methods ignore the significant individual differences between subjects, making it difficult for models trained based on some subjects to perform well on new subjects.

B. Graph Convolutional Network

Graph convolutional network (GCN) is a powerful technique for handling graph-structured data. It learns node representations by aggregating neighborhood information using neighbor transfer functions. Given the striking similarity between EEG signals and graph data, GCN-based methods have rapidly advanced in EEG-based emotion recognition. For instance, Song et al. [29] introduced a dynamic GCN (DGCN) method that explores the intrinsic relationship between EEG channels by learning the global adjacency matrix. Zhong et al. [30] extended DGCN with regularization to propose RGNN. Ye et al. [15] explored multi-level spatial information between EEG channels and introduced a hierarchical dynamic GCN (HD-GCN). Jin et al. [31] introduced the pyramidal graph convolutional network (PGCN) method, aggregating local, mesoscopic, and global features to enhance brain function modeling. While these GCN-based methods tend to learn global intrinsic relationships between different EEG channels for generalizable features, they overlook class-level substructures within different categories, resulting in limited performance.

C. EEG-Based Domain Adaptation Emotion Recognition

DA aims to learn domain-invariant features to transfer knowledge from the source domain to the target domain. In computer vision, different domains may refer to images with different styles and types. For EEG signals, different domains tend to represent different subjects. Several DA methods have been employed to alleviate individual differences. For instance, Li et al. [32] introduced a bi-hemisphere domain adversarial neural network (BiDANN) by training domain discriminators to reduce domain differences in each hemisphere between the source and target domains. Yang et al. [22] proposed a spatial-spatial attention alignment multi-source domain adaptation (SSAA-MSDA) approach to minimize affective cognitive pattern discrepancies among subjects to alleviate the differences in domain feature distribution. However, most existing DA methods focus on minimizing domain marginal distribution discrepancies, overlooking class-wise conditional distributions between domains, resulting in insufficient discriminative domain-invariant features. Recently, some approaches have attempted to align class-wise semantic distributions to improve model transferability. For example, Li et al. [20] proposed an unsupervised dynamic domain adaptation (UDDA) method to obtain target semantics through a pseudo-labeling method based on individual point thresholds to align intra-class subdomains. Zhou et al. [21] proposed a prototypical representation-based pairwise learning (PR-PL) method that dynamically filtered unreliable samples and employed pairwise learning to capture the proximity relationships. However, these methods only consider

the knowledge of individual points and ignore the semantic proximity relationship between the source and target samples, resulting in noisy semantic information or negative transfer.

III. PROPOSED METHOD

In a DA task, we typically have source domains with emotion labels, denoted as $\mathcal{D}_s = \{x_i^s, y_i^s\}_{i=1}^n$, and unlabeled target domain, denoted as $\mathcal{D}_t = \{x_j^t, y_j^t\}_{j=1}^m$. Here, x_i^s and x_j^t represent EEG samples, while y_i^s and y_j^t correspond to the respective emotion labels. A typical assumption is made that the distribution of EEG signals differs between the source domains and the target domain, i.e., $\mathcal{P}(x_i^s) \neq \mathcal{P}(x_j^t)$. Moreover, there may also exist a discrepancy in the conditional distribution between the source and target domains, i.e., $\mathcal{P}(y_i^s | x_i^s) \neq \mathcal{P}(y_j^t | x_j^t)$.

A. Preprocessing and Feature Extraction

In the preprocessing stage, we downsampled the EEG signals to 200 Hz and applied a band-pass filter to eliminate noise and power frequency interference. Subsequently, we used independent component analysis (ICA) to remove various artifacts from the EEG signals, including those caused by head movement, electrooculography, and eye drift, which enhances the quality and reliability of EEG signals. For feature extraction, we utilized the differential entropy (DE) feature to characterize the emotional features because of their stability and distinguishability. We calculated the DE features across five frequency bands, including δ band (1-3 Hz), θ band (4-7 Hz), α band (8-13 Hz), β band (14-30 Hz), and γ band (31-50 Hz).

Assuming that the EEG data follows the Gaussian distribution, the DE features can be defined as:

$$h(x) = - \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma^2} \exp \frac{(x-\mu)^2}{2\sigma^2} \log \frac{1}{\sqrt{2\pi}\sigma^2} \exp \frac{(x-\mu)^2}{2\sigma^2} \\ = \frac{1}{2} \log 2\pi e \sigma^2 \quad (1)$$

where x is the EEG signals, μ , and σ is the mean and variance of x .

We segmented the EEG signals at 1-second intervals so that each trial had between 185 and 238 samples. Each EEG sample comprises 62 channels and five frequency bands, and the feature dimension of each segment is 310 (62×5).

B. Graph Attention Convolutional Network

The intricate connectivity patterns among electrodes and brain regions play a crucial role in understanding brain dynamics and emotional processes in EEG signals. Therefore, it is crucial that the model can learn the affective cognition function in EEG-based emotion recognition. As shown in Fig. 2, we introduce a DGAT method for dynamically learning the global and local class-wise intrinsic relationships between brain regions, providing valuable insights into complex connectivity patterns.

1) *Graph Convolutional Network*: Consider an undirected graph denoted as $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ represents a set of nodes (EEG channels) and $\mathcal{E}$ denotes a set of edges connecting nodes within $\mathcal{V}$. For an EEG sample, the input is denoted

(R2G-STNN)方法，能够从局部到全局区域进行层次化特征学习，以学习判别性时空特征。尽管这些方法表现出色，但它们忽略了受试者之间的显著个体差异，导致基于某些受试者训练的模型难以在新受试者上表现良好。

B. 图卷积网络

图卷积网络（GCN）是一种处理图结构数据的强大技术。它通过使用邻域传递函数聚合邻域信息来学习节点表示。鉴于脑电图（EEG）信号与图数据之间的显著相似性，基于 GCN 的方法在基于 EEG 的情绪识别领域迅速发展。例如，Song 等人[29]引入了一种动态 GCN（DGCN）方法，该方法通过学习全局邻接矩阵来探索 EEG 通道之间的内在关系。Zhong 等人[30]通过正则化扩展了 DGCN，提出了 RGNN。Ye 等人[15]探索了 EEG 通道之间的多级空间信息，并引入了一种分层动态 GCN（HD-GCN）。Jin 等人[31]引入了金字塔图卷积网络（PGCN）方法，通过聚合局部、中观和全局特征来增强脑功能建模。虽然这些基于 GCN 的方法倾向于学习不同 EEG 通道之间的全局内在关系以获得泛化特征，但它们忽略了不同类别内的类别级子结构，导致性能有限。

C. 基于 EEG 的领域自适应情绪识别

领域自适应（DA）旨在学习领域不变特征，以将知识从源域迁移到目标域。在计算机视觉中，不同的领域可能指具有不同风格和类型的图像。对于脑电图（EEG）信号，不同的领域往往代表不同的被试者。已经采用了几种 DA 方法来减轻个体差异。例如，Li 等人[32]通过训练领域判别器来引入一个双半球领域对抗神经网络（BiDANN），以减少源域和目标域在每个半球中的领域差异。Yang 等人[22]提出了一种空间-空间注意力对齐多源领域自适应（SSAA-MSDA）方法，以最小化被试者之间的情感认知模式差异，从而减轻领域特征分布的差异。然而，大多数现有的 DA 方法专注于最小化领域边缘分布差异，忽略了领域之间的类别条件分布，导致领域不变特征区分不足。最近，一些方法尝试对齐类别语义分布，以提高模型迁移能力。例如，Li 等人。[20]提出了一种无监督动态领域自适应（UDDA）方法，通过基于个体点阈值的伪标签方法来获取目标语义，以对齐类内子域。Zhou 等人[21]提出了一种基于原型表示的成对学习（PR-PL）方法，该方法动态过滤不可靠样本，并采用成对学习来捕获邻近关系。然而

这些方法仅考虑单个点的知识，忽略了源样本和目标样本之间的语义邻近关系，导致语义信息噪声或负迁移。

III. PM 在领域自适应任务中，我们通常有带有情绪标签的源域，记为 $D = \{x, y\}$ ，以及未标记的目标域，记为 $D = \{x, y\}$ 。这里， x 和 x 表示脑电图样本，而 y 和 y 对应相应的情绪标签。一个典型假设是脑电信号的分布在不同源域和目标域之间有所不同，即 $P(x) \neq P(x)$ 。此外，源域和目标域之间的条件分布也可能存在差异，即 $P(y|x) \neq P(y|x)$ 。

A. 预处理和特征提取

在预处理阶段，我们将脑电图（EEG）信号降采样至 200 Hz，并应用带通滤波器以消除噪声和工频干扰。随后，我们使用独立成分分析（ICA）去除脑电图信号中的各种伪影，包括由头部运动、眼电图和眼漂移引起的伪影，这提高了脑电图信号的质量和可靠性。对于特征提取，我们利用微分熵（DE）特征来表征情绪特征，因为它们具有稳定性和可区分性。我们在五个频段上计算 DE 特征，包括 δ 频段（1-3 Hz）、 θ 频段（4-7 Hz）、 α 频段（8-13 Hz）、 β 频段（14-30 Hz）和 γ 频段（31-50 Hz）。

假设脑电图数据服从高斯分布，DE 特征可以定义为：

$$\begin{aligned} h(x) &= - \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma}} \exp\left(-\frac{(x-\mu)^2}{2\sigma}\right) \log \frac{1}{\sqrt{2\pi\sigma}} \exp\left(-\frac{(x-\mu)^2}{2\sigma}\right) dx \\ &= \frac{1}{2} \log 2\pi e \sigma \end{aligned} \tag{1}$$

其中 x 是脑电图信号， μ 和 σ 分别是 x 的均值和方差。

我们将脑电图信号以 1 秒的间隔进行分段，以便每个试验包含 185 到 238 个样本。每个脑电图样本包含 62 个通道和五个频段，每个分段的特征维度为 310（ 62×5 ）。

B. 图注意力卷积网络

电极和脑区之间的复杂连接模式在理解脑动力学和脑电信号中的情绪过程方面起着至关重要的作用。因此，模型能够学习脑电情感识别中的情感认知功能至关重要。如图 2 所示，我们引入了一种 DGAT 方法，用于动态学习脑区之间的全局和局部类别内在关系，为复杂的连接模式提供了宝贵的见解。

- 1) 图卷积网络：考虑一个无向图表示为 $G = (V, E)$ ，其中 V 表示节点集（EEG 通道）， E 表示连接 V 内节点的边集。对于 EEG 样本，输入表示为

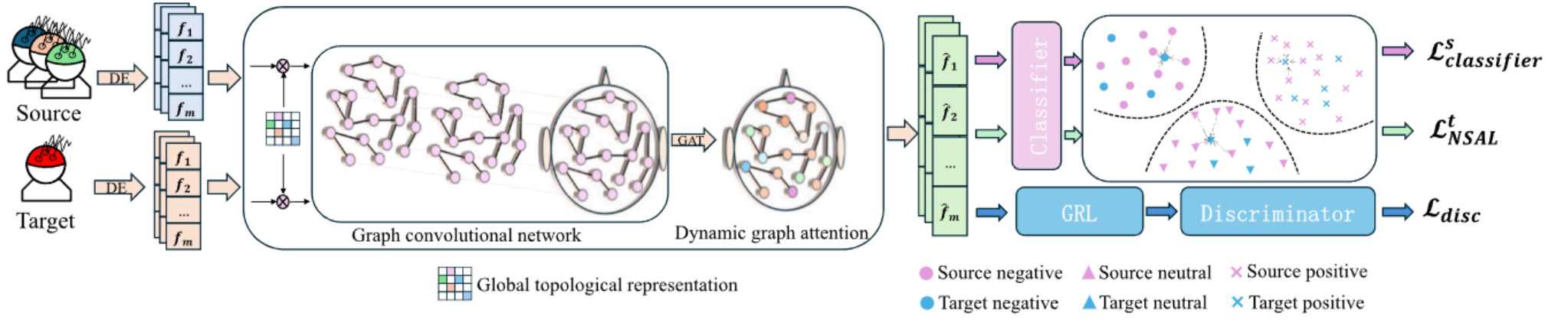


Fig. 2. The overall framework of NSAL-DGAT. The NSAL-DGAT consists of a dynamic graph attention convolutional neural network (DGAT) and neighborhood semantic aware learning (NSAL). The DGAT extracts the global and local class-wise graph representation to learn discriminative features. The NSAL aims to learn target semantics through neighborhood source samples and incorporate them as a regularization for improving model adaptation.

as $\mathbf{X} \in \mathbb{R}^{C \times B}$, where $C = |\mathcal{V}|$ corresponds to the number of EEG channels and B denotes the DE features extracted from different frequency bands. The edge set $\mathcal{E}$ can be represented by a weighted adjacency matrix $\mathbf{A} \in \mathbb{R}^{C \times C}$, where the (i, j) -th element quantifies the coupling strength between the i -th and j -th EEG channels. In general, a fixed adjacency matrix $\mathbf{A}$ is computed based on the physical distances or similarities between nodes, establishing strong local connections between nodes that are either spatially close or highly similar. However, it is necessary to learn the global brain functional structure. Therefore, we introduce the dynamic adjacency matrix $\mathbf{A}^G$, which can be defined as:

$$\mathbf{A}^G = \sigma(W_2 \delta(W_1 \mathbf{A})) \quad (2)$$

where $\mathbf{A} \in \mathbb{R}^{(C \times C)}$ is a randomly initialized dense adjacency matrix, $W_1 \in \mathbb{R}^{(\frac{C \times C}{r}) \times (C \times C)}$ and $W_2 \in \mathbb{R}^{(C \times C) \times (\frac{C \times C}{r})}$ are weight matrices, $\mathbf{A}^G \in \mathbb{R}^{C \times C}$ is vectorized by $\mathbf{A}$, σ and δ are the ELU and Tanh functions, and r is the reduction ratio. First, we apply a linear mapping to the dense adjacency matrix $\mathbf{A}$, followed by the ReLU function to obtain the non-negative adjacency matrix $\mathbf{A}^G$. Here, the learnable parameter A_{ij}^G denotes the coupling strength between the i -th and j -th EEG channels.

We employ the simple GCN layer [33] to learn the global brain functional structure based on the obtained learnable adjacency matrix $\mathbf{A}^G$. The m -th GCN is defined as follows:

$$\mathbf{Z}_m = \delta_2(D^{-1} \mathbf{A}^G \delta_1(\mathbf{Z}_{m-1} \mathbf{W}_{m1}) \mathbf{W}_{m2} + \mathbf{Z}_{m-1}) \quad (3)$$

where $\mathbf{Z}_{m-1}, \mathbf{Z}_m \in \mathbb{R}^{(C \times B)}$ are input and output node representation at m -th GCN layer, respectively. The initial input feature, denoted as $\mathbf{Z}_0$, corresponds to the original input feature $\mathbf{X}$. The matrix D^{-1} represents the degree matrix of the adjacency matrix $\mathbf{A}^G$. The learnable weight matrixes are $\mathbf{W}_{m1} \in \mathbb{R}^{(B \times B')}$, $\mathbf{W}_{m2} \in \mathbb{R}^{(B' \times B)}$, where B serves as an adjustable hyperparameter, and δ_1 and δ_2 denote ELU functions.

To avoid GCN over-smoothing, we introduce a two-layer GCN network with a residual architecture. The final aggregated features can be expressed as:

$$\mathbf{Z} = \text{concat}(\mathbf{X}, \mathbf{Z}_1, \mathbf{Z}_2) \quad (4)$$

2) *Dynamic Graph Attention*: Recent GCN methods have primarily focused on learning the global intrinsic relationships among EEG channels. However, they often overlook class-wise substructures associated with different emotional states. This

oversight may hinder the learning of generalizable and discriminative representations. To mitigate this limitation, we propose a simple and effective approach, dynamic graph attention (DGAT). The DGAT enables the model to learn crucial node features for different emotions dynamically.

The DGAT incorporates an attention mechanism that selectively emphasizes spatial information (i.e., EEG channels) to identify critical features. Specifically, the input to DGAT is EEG features represented as $\mathbf{Z} \in \mathbb{R}^{C \times S}$, where S corresponds to spectral information, and C refers to spatial information. The module aggregates feature information by employing both average and max pooling operations, resulting in two one-dimensional vectors: F_{avg} and F_{max} . Subsequently, these vectors are fed into a shared multilayer perceptron (MLP) with a single hidden layer. The MLP compresses the feature dimensions and combines them element-wise to generate the attention matrix $\mathbf{M}_Z$. The attention mechanism [22] is defined as follows:

$$\begin{aligned} \mathbf{M}_Z &= \sigma(\text{MLP}(\text{AvgPool}(\mathbf{Z})) + \text{MLP}(\text{MaxPool}(\mathbf{Z}))) \\ &= \sigma(W_1 \mathbf{W}_0 \mathbf{Z}_{avg} + W_1 \mathbf{W}_0 \mathbf{Z}_{max}) \end{aligned} \quad (5)$$

where $\mathbf{Z}_{avg} \in \mathbb{R}^C$ and $\mathbf{Z}_{max} \in \mathbb{R}^C$ are the channel features after average pooling and maximum pooling, $\mathbf{W}_0 \in \mathbb{R}^{\frac{C}{r} \times C}$ and $\mathbf{W}_1 \in \mathbb{R}^{C \times \frac{C}{r}}$ are the weights of the MLP. The σ is an activation function to normalize the attention weight between 0 and 1, and r is an adjustable hyperparameter. Finally, we obtain the final discriminative features $\mathbf{F}$:

$$\mathbf{F} = \mathbf{Z} + \mathbf{Z} \mathbf{M}_Z \quad (6)$$

C. Neighborhood Semantic Aware Learning

Most existing DA methods alleviate individual differences by aligning the global data distribution between the source and target domains. However, these methods overlook the class-wise semantic information of the target domain, resulting in domain-invariant features that are limited in their discriminative ability to distinguish between different emotional states. Here, we assume that the instances from the same class in the source and target domains will be clustered together after the domain adaptation. Based on this assumption, we introduce neighborhood semantic-aware learning to learn semantic information of the target domain and encourage class-level feature distribution alignment.

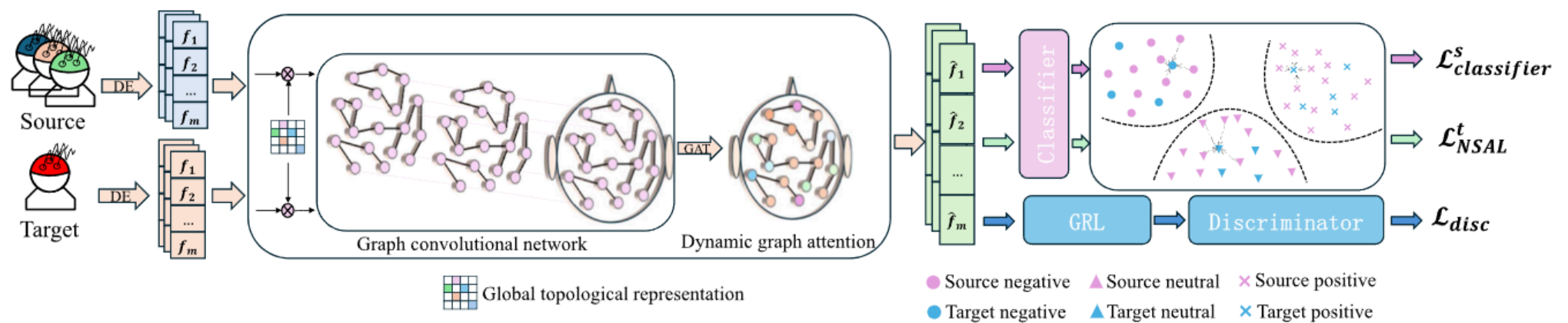


图 2. NSAL-DGAT 的整体框架。NSAL-DGAT 由一个动态图注意力卷积神经网络 (DGAT) 和邻域语义感知学习 (NSAL) 组成。DGAT 提取全局和局部的类别图表示以学习判别性特征。NSAL 旨在通过邻域源样本学习目标语义, 并将其作为正则化以改进模型自适应。

当 X 属于实数集 R 时, 其中 C 等于 $|V|$, 对应于脑电图 (EEG) 通道的数量, B 表示从不同频段提取的深度特征 (DE) 特征。边集 E 可以表示为加权邻接矩阵 A 属于 R , 其中 (i, j) 元素量化了第 i 个和第 j 个 EEG 通道之间的耦合强度。通常, 基于节点之间的物理距离或相似性计算固定的邻接矩阵 A , 建立空间上接近或高度相似的节点之间的强局部连接。然而, 有必要学习全局脑功能结构。因此, 我们引入动态邻接矩阵 A , 可以定义为:

$$A = \sigma(W\delta(WA)) \quad (2)$$

其中 A 属于 R 是一个随机初始化的密集邻接矩阵, W 属于 R 和 W 属于 R 是权重矩阵, A 属于 R 通过 A 进行向量化, σ 和 δ 是 ELU 和 Tanh 函数, r 是缩减率。首先, 我们对密集邻接矩阵 A 应用线性映射, 然后通过 ReLU 函数得到非负邻接矩阵 A 。这里, 可学习参数 A 表示第 i 个和第 j 个 EEG 通道之间的耦合强度。

我们采用简单的 GCN 层[33]基于获得的可学习邻接矩阵 A 来学习全局脑功能结构。第 m 个 GCN 定义为如下:

$$Z = \delta \left(DA\delta(ZW)W + Z \right) \quad (3)$$

其中 Z 和 $Z \in R$ 分别是第 m 个 GCN 层的输入和输出节点表示。初始输入特征用 Z 表示, 对应于原始输入特征 X 。矩阵 D 表示邻接矩阵 A 的度矩阵。可学习权重矩阵 $W \in R$, $W \in R$, 其中 B 作为可调节的超参数, δ 和 δ 表示 ELU 函数。

为了避免图卷积网络 (GCN) 过度平滑, 我们引入了一个具有残差结构的双层 GCN 网络。最终聚合的特征可以表示为:

$$Z = \text{concat}(X, Z, Z) \quad (4)$$

2) 动态图注意力: 最近的 GCN 方法有主要集中于学习 EEG 通道之间的全局内在关系。然而, 它们常常忽略了与不同情绪状态相关的类别级子结构

这种疏忽可能会阻碍学习具有泛化性和区分性的表示。为了减轻这一局限性, 我们提出了一种简单而有效的方法, 即动态图注意力 (DGAT)。DGAT 使模型能够动态地学习不同情绪的关键节点特征。

DGAT 结合了一种注意力机制, 该机制选择性地强调空间信息 (即 EEG 通道) 以识别关键特征。具体来说, DGAT 的输入是表示为 $Z \in R$ 的 EEG 特征, 其中 S 对应于频谱信息, C 指空间信息。该模块通过采用平均池化和最大池化操作来聚合特征信息, 从而得到两个一维向量: F 和 F 。随后, 这些向量被输入到一个具有单个隐藏层的共享多层感知器 (MLP)。MLP 压缩特征维度并将它们逐元素组合以生成注意力矩阵 M 。注意力机制 [22] 定义如下:

$$\begin{aligned} M &= \sigma(MLP(AvgPool(Z)) + MLP(MaxPool(Z))) \\ &= \sigma(WWZ + WWZ) \end{aligned} \quad (5)$$

其中 $Z \in R$ 和 $Z \in R$ 分别是经过平均池化和最大池化后的通道特征, $W \in R$ 和 $W \in R$ 分别是 MLP 的权重。 σ 是一个激活函数, 用于将注意力权重归一化到 0 和 1 之间, r 是一个可调节的超参数。最后, 我们得到最终的判别性特征 F :

$$F = Z + ZM \quad (6)$$

C. 邻近语义感知学习

大多数现有的域适应 (DA) 方法通过对齐源域和目标域的全局数据分布来缓解个体差异。然而, 这些方法忽略了目标域的类别级语义信息, 导致域不变特征在区分不同情绪状态方面的判别能力有限。在这里, 我们假设在域适应后, 源域和目标域中属于同一类别的实例将被聚集在一起。基于这一假设, 我们引入邻近语义感知学习来学习目标域的语义信息, 并鼓励类别级特征分布对齐。

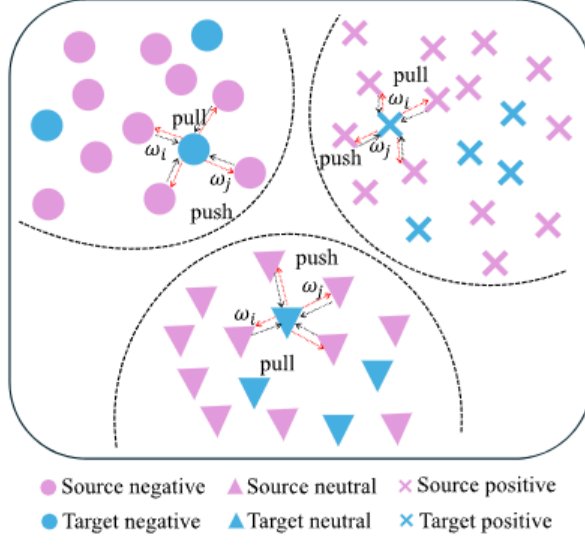


Fig. 3. The entropy-weighted strategy encourages the target instances toward the source individuals with low entropy in the neighborhood of source samples.

Specifically, we assume that target samples and nearest neighbor source samples share the same emotion labels after domain adaptation. To identify the semantically closest neighbors [34], we construct a feature library $\mathcal{F} = G_f(x_s)$ to store source domain features, where $G_f(\cdot)$ represents the DGAT feature extractor that contains two layers of GCN and dynamic graph attention. Next, we build a score library $S = (G_y(G_f(x_s)))$ to store the corresponding prediction scores, which $G_y(\cdot)$ is responsible for distinguishing the emotion category. To find the k nearest neighbor source domain features $\mathcal{N}$ for each target instance in the feature library $\mathcal{F}$, we use the cosine similarity, as shown in the following formula:

$$\mathcal{N}_{\{1, \dots, K\}} = \{\mathcal{F}_j | \text{top-}K(\cos(f(x_i), f(x_j)), \forall x_j \in D_s)\} \quad (7)$$

Since target domain samples near the decision boundary are prone to misclassification, this can impact the accuracy of the obtained semantic information. However, source domain samples with low entropy values represent high-confidence samples that are easily distinguishable by the classifier $G_y(\cdot)$. As shown in Fig. 3, unlike previous methods that rely on the mean value to obtain semantic information from the target domain, the proposed NSAL introduces an entropy-weighted strategy. This strategy calculates the weights of the k nearest neighbor source domain features to ensure that the target domain samples are positioned further from the decision boundary and closer to the source samples with low entropy. The specific formula for this strategy is as follows:

$$\bar{p} = \frac{1}{K} \sum_{j=1}^K w_j \cdot p_c(x_j) \quad (8)$$

where w_j denotes the weight of j -th nearest neighbor source domain features and can be calculated using the entropy method, as follows:

$$w_j = \frac{1 - (-\frac{1}{K} (\sum_{i=1}^K p_c(x_{ij}) \ln p_c(x_{ij})))}{\sum_{j=1}^K (1 - (-\frac{1}{K} (\sum_{i=1}^K p_c(x_{ij}) \ln p_c(x_{ij})))} \quad (9)$$

where $p_c(x_{ij})$ represents the predicted value of the i -th emotional category corresponding to the j -th nearest neighbor source domain feature in the score library S , and w_j represents the

weight assigned to it by the entropy value of the j th nearest neighbor.

Then, we leverage the semantic target samples to improve the transferability of the model. This method learns the target semantics to help align the class-wise conditional distributions between domains, thus acquiring discriminative domain-invariant features. Specifically, we can use the following formula to express:

$$\mathcal{L}_{\text{NSAL}}^t = -\frac{1}{n} \sum_{i=1}^n y_i \log P_{\theta}(\hat{y}_i | \mathbf{x}_i) \quad (10)$$

here, y_i represents the pseudo labels of target domain data from k nearest neighbor source domain features, n is the total number of samples, and $P_{\theta}(\hat{y}_i | \mathbf{x}_i)$ denotes the probability distribution $\hat{y}_i$.

D. Optimizing NSAL-DGAT

As shown in Fig. 2, the proposed NSAL-DGAT employs adversarial training to minimize domain differences and integrate target semantics as a regularization to alleviate individual differences in EEG signals and improve the generalization ability of the model. Specifically, in NSAL-DGAT, the DGAT feature extractor G_f is used to extract high-dimensional features. Then, a domain discriminator G_d is designed to reduce the marginal distribution difference across domains. Furthermore, NSAL is introduced as a regularization method to learn target semantic information and encourage class-wise conditional probability distribution alignment to obtain discriminative domain-invariant features. Then, we will introduce how to optimize the loss function in NSAL-DGAT.

1) *Label Classifier*: The label classifier (G_y) is used to distinguish the labels of the input features. Here, the source domain data D_s is used for supervised learning, and the feature classifier G_f and the label classifier G_y are trained by minimizing the cross-entropy loss, which can be expressed as:

$$\mathcal{L}_{\text{classifier}}^s = -\frac{1}{n} \sum_{i=1}^n y_i \log P_{\theta}(\hat{y}_i | \mathbf{x}_i) \quad (11)$$

where y_i is the true labels of source domain data, n is the number of samples, and $P_{\theta}(\hat{y}_i | \mathbf{x}_i)$ denotes the probability distribution of predicted label $\hat{y}_i$.

2) *Domain Discriminator*: The domain discriminator (G_d) [22], [35] minimizes the marginal (global) distribution between domains through adversarial training, which is given as:

$$\mathcal{L}_{\text{disc}} = \frac{1}{n_s + n_t} \sum_{\mathbf{x}_i \in \mathcal{D}_s \cup \mathcal{D}_t} L_d(G_d(G_f(\mathbf{x}_i)), d_i) \quad (12)$$

where $\mathcal{L}_{\text{disc}}$ denotes the standard two-category cross-entropy loss. For input samples originating from the source domain, d_i takes the value of 1, while for input samples originating from the target domain, d_i is set to 0.

The overall loss function of the proposed NSAL-DGAT (13) can be expressed as the supervised cross-entropy loss on the source domain (11), the domain discriminator loss (12), and the

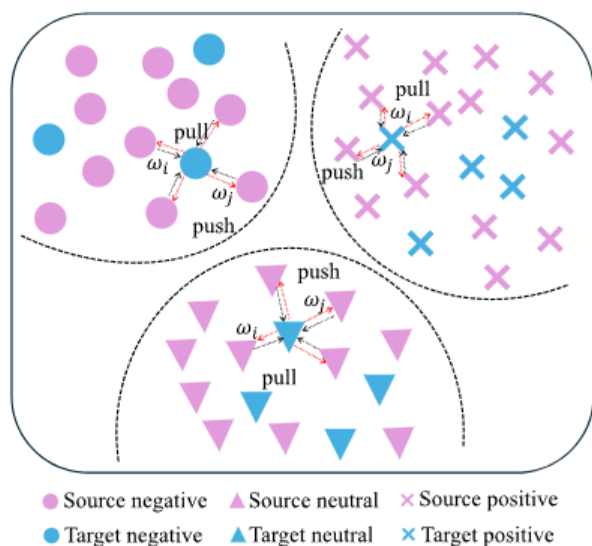


图 3. 熵权策略鼓励目标实例向源样本邻域中熵值低的源个体靠拢。

具体而言，我们假设在领域自适应后，目标样本和最近邻源样本共享相同的情绪标签。为了识别语义上最接近的邻居[34]，我们构建了一个特征库 $F = G(x)$ 来存储源域特征，其中 $G(\cdot)$ 表示包含两层 GCN 和动态图注意力的 DGAT 特征提取器。接下来，我们构建了一个分数库 $S = (G(G(x)))$ 来存储相应的预测分数， $G(\cdot)$ 负责区分情绪类别。为了在特征库 F 中为每个目标实例找到 k 个最近邻源域特征 N ，我们使用余弦相似度，如下公式所示：

$$N = \{F | \text{top-}K(\cos(f(x), f(x)), \forall x \in D)\} \quad (7)$$

由于目标域中靠近决策边界的样本容易发生误分类，这会影响所获得语义信息的准确性。然而，熵值较低的源域样本代表高置信度样本，这些样本能够被分类器 $G(\cdot)$ 轻易区分。如图 3 所示，与以往依赖均值从目标域获取语义信息的方法不同，所提出的 NSAL 引入了一种熵加权策略。该策略计算 k 个最近邻源域特征的权重，以确保目标域样本远离决策边界，更接近熵值较低的源域样本。该策略的具体公式如下：

$$\bar{p} = \frac{1}{K} \sum_{j=1}^K w \cdot p(x) \quad (8)$$

其中 w 表示第 j 个最近邻源域特征的权重，可以使用熵权法计算，如下：

$$w = \frac{\sum_{i=1}^K \frac{1 - (-\sum_{i=1}^K p(x) \ln p(x))}{1 - (-\sum_{i=1}^K p(x) \ln p(x))}}{\sum_{i=1}^K \frac{1 - (-\sum_{i=1}^K p(x) \ln p(x))}{1 - (-\sum_{i=1}^K p(x) \ln p(x))}} \quad (9)$$

其中， $p(x)$ 表示在得分库 S 中与第 j 个最近邻源域特征对应的第 i 个情绪类别的预测值)

并且 w 表示由第 j 个最近邻的熵值分配给它的权重。

然后，我们利用语义目标样本来提高模型的迁移能力。这种方法学习目标语义以帮助对齐不同领域之间的类别条件分布，从而获得具有区分性的领域不变特征。具体来说，我们可以使用以下公式来表示：

$$L = - \frac{1}{n} \sum_{i=1}^n y \log P(\hat{y} | x) \quad (10)$$

这里， y 表示从 k 个最近邻源域特征中得到的伪标签， n 是样本总数， $P(\hat{y} | x)$ 表示概率分布

$\hat{y}$ 。

D. 优化 NSAL-DGAT

如图 2 所示，所提出的 NSAL-DGAT 采用对抗训练来最小化域差异，并将目标语义作为正则化来减轻脑电图信号中的个体差异，提高模型的泛化能力。具体来说，在 NSAL-DGAT 中，DGAT 特征提取器 G 用于提取高维特征。然后，设计了一个域判别器 G 来减少跨域的边际分布差异。此外，引入 NSAL 作为正则化方法来学习目标语义信息，并鼓励类别条件概率分布对齐，以获得具有域不变性的判别特征。然后，我们将介绍如何优化 NSAL-DGAT 的损失函数。

1) 标签分类器：标签分类器 (G) 用于区分输入特征的标签。这里，使用源域数据 Dis 进行监督学习，特征分类器 G 和标签分类器 G 通过最小化交叉熵损失进行训练，可以表示为：

$$L_{\text{classifier}} = - \frac{1}{n} \sum_{i=1}^n y \log P(\hat{y} | x) \quad (11)$$

其中 y 是源域数据的真实标签， n 是样本数量， $P(\hat{y} | x)$ 表示预测标签 $\hat{y}$ 的概率分布。

2) 域判别器：域判别器 (G) [22], [35] 通过对抗训练最小化域之间的边缘（全局）分布，其表达式为：

$$L = \frac{1}{n+n} \sum_{x \in D \cup D} L(G(G(x)), d) \quad (12)$$

其中 L 表示标准的二分类交叉熵损失。对于来自源域的输入样本， d_i 取值为 1，而对于来自目标域的输入样本， d_i 设置为 0。

所提出的 NSAL-DGAT (13) 的整体损失函数可以表示为源域的监督交叉熵损失 (11)、域判别器损失 (12) 以及

NSAL regularization loss on the target domain (10) is given as:

$$\min_{\theta_f} \max_{\theta_{disc}} \mathcal{L}_{\text{classifier}}^s + \mathcal{L}_{disc} + \lambda \mathcal{L}_{\text{NSAL}}^t \quad (13)$$

Here, the hyperparameter γ [21] regulates the significance of the NSAL loss and is defined as:

$$\gamma = \delta \times \frac{\text{epochidx}}{\text{maxepoch}} \quad (14)$$

where *epochidx* represents the current epoch index, while *maxepoch* signifies the total number of training epochs. The δ is a constant value.

IV. EXPERIMENTS AND RESULTS

This section presents the results of our experiments. Specifically, Section IV-A introduces the SEED and HIED datasets; Section IV-B describes the experimental settings, both subject-dependent and subject-independent scenarios; Section IV-C introduces the model structure and experimental parameters; Section IV-D presents the experimental results; and Section IV-E shows the ablation experiments.

A. Datasets

SEED: This dataset comprises EEG signals elicited by Chinese movie clips [36], encompassing three emotions: positive, neutral, and negative. It contains 15 movie clips, with 5 clips per emotion category, each lasting 4 minutes. These EEG signals were collected from 15 subjects (7 males and 8 females), with each subject taking part in three experiments conducted at different times, resulting in a total of 45 experiments. After signal preprocessing, the EEG signals are segmented into 1-s intervals, leading to a total of 3394 samples per experiment.

HIED: This dataset, collected from Technical College for The individuals with hearing loss at Tianjin University of Technology [37], [38], focuses on EEG-based emotion recognition and explores the impact of hearing impairment on affective cognitive functions. Similar to the SEED dataset, this dataset utilizes identical movie clip stimuli to elicit EEG signals associated with three emotions: positive, neutral, and negative. The experiment consists of 15 trials, with 5 trials per emotional category. Each trial follows a structured protocol: a 5-second hint precedes the stimulus, followed by a 45-second self-evaluation period where subjects rate their subjective feelings, and then a 15-second rest period to allow for state adjustment. The dataset comprises EEG signals recorded from 15 hearing-impaired volunteers (3 males and 12 females) aged between 17 and 24 years (mean: 21 years, standard deviation: 2 years). The EEG signals were recorded using the NeuroScan system at a sampling rate of 1000 Hz, with an electrode cap containing 62 channels. Subsequently, the EEG signals were segmented into 1-second segments, and DE features were extracted from five frequency bands. Thus, each experiment comprises a total of 3433 samples.

B. Experimental Protocol

To comprehensively evaluate the performance of the NSAL-DGAT method, we conducted the well-known experimental protocols from prior works [21], [29], [32] on the SEED and HIED datasets, including subject-dependent and subject-independent experiments. (1) *Subject-dependent experiment.* For each subject in one session, the first 9 trials are used as the source domain, and the remaining 6 trials serve as the target domain. The final results are obtained by calculating the average accuracy across subjects. (2) *Subject-independent experiment.* Here, we utilize the widely used leave-one-subject-out strategy. Data from one session of a single subject are treated as the target domain, while data from a single session of the remaining subjects form the source domain. For example, in the SEED dataset, each subject has 3394 samples in one session, leading to a source domain size of 14×3394 and a target domain size of 1×3394 . We repeated this process 15 times to obtain results for each subject, and the average performance across all subjects is reported as the final results. Here, as in previous studies [21], the SEED dataset contains three sessions. The first session represents the initial viewing of the stimulus material, during which emotional feedback tends to be stronger compared to the subsequent sessions. Therefore, our analysis focuses on the first session in both subject-dependent and subject-independent scenarios.

C. Implementation Results

The proposed NSAL-DGAT method consists of two main components: a dynamic graph attention feature extractor (DGAT), referred to as G_f , and a feature discriminator, denoted as G_d . The DGAT feature extractor, G_f , comprises a two-layer GCN and a graph attention network. The feature discriminator, G_d , is implemented as a multi-layer perceptron (MLP) with layer sizes of 64 - 64 - 2 and employs a Rectified Linear Unit (ReLU) activation function. Furthermore, the RMSprop optimizer is utilized for network training with a learning rate set to 0.001 and a batch size of 48. To mitigate overfitting, L_2 regularization ($1e-3$) is applied in the network. These experiments are executed on an NVIDIA GeForce RTX 2080 GPU, using CUDA 10.0 and PyTorch for efficient computation and deep learning operations.

D. Experimental Results

In this section, we conduct extensive experiments on two datasets for two scenarios. The experimental results are described as follows.

1) *Subject-Dependent Experiment on SEED:* To evaluate the effectiveness of the proposed NSAL-DGAT method, we use a subject-dependent protocol and carefully select two categories of representative baseline methods to conduct extensive experiments. The first group comprises traditional methods, including SVM [39], RF [40], CCA [41], GRSLR [25], GSCCA [26] and deep belief networks (DBN) [36]. The other group comprises deep learning models, namely GCNN [14], DANN [17], DGCNN [29], BIHDM [42], RGNN [30], BiDANN [32], HD-GCN [15], and PR-PL [21].

目标域上的 NSAL 正则化损失 (10) 给出为：

$$\min_{\theta} \max_{\theta} L_{\text{classifier}} + L + \lambda L \quad (13)$$

这里，超参数 γ [21] 调节 NSAL 损失的重要性，定义为：

$$\gamma = \delta \times \frac{\text{epochidx}}{\text{maxepoch}} \quad (14)$$

where epochidx 表示当前周期索引，而 maxepoch 代表训练周期的总数。 δ 是一个常数值。

IV. ER 本节介绍了我们的实验结果。具体来说，第 IV-A 节介绍了 SEED 和 HIED 数据集；第 IV-B 节描述了实验设置，包括受试者相关和受试者无关的场景；第 IV-C 节介绍了模型结构和实验参数；第 IV-D 节展示了实验结果；第 IV-E 节展示了消融实验。

A. 数据集

SEED：该数据集包含由中国电影片段诱发的脑电图信号 [36]，涵盖三种情绪：积极、中性、消极。它包含 15 个电影片段，每种情绪类别有 5 个片段，每个片段持续 4 分钟。这些脑电图信号从 15 名受试者（7 名男性，8 名女性）收集，每位受试者在不同时间参与三次实验，总共 45 次实验。经过信号预处理后，脑电图信号被分割成 1 秒的间隔，每个实验产生 3394 个样本。HIED：该数据集采集自天津工业大学听力障碍人士技术学院 [37], [38]，专注于基于脑电图的情绪识别，并探讨了听力障碍对情感认知功能的影响。与 SEED 数据集类似，该数据集使用相同的电影片段刺激来诱发与三种情绪（积极、中性、消极）相关的脑电信号。实验包含 15 次试验，每类情绪 5 次试验。每次试验遵循结构化协议：刺激前有 5 秒提示，随后是 45 秒的自我评估期，受试者在此期间评价其主观感受，然后是 15 秒的休息期以进行状态调整。该数据集包含 15 名听力障碍志愿者（3 名男性，12 名女性）录制的脑电信号，年龄在 17 至 24 岁之间（平均年龄 21 岁，标准差 2 岁）。脑电信号使用 NeuroScan 系统以 1000 Hz 的采样率记录，使用包含 62 个通道的电极帽。随后，脑电信号被分割为 1 秒的片段，并从五个频段中提取 DE 特征。因此，每个实验总共包含 3433 个样本。

B. 实验协议

为了全面评估 NSALDGAT 方法的表现，我们在 SEED 和 HIED 数据集上进行了先前工作中的著名实验协议[21]、[29]、[32]，包括受试者相关和受试者无关的实验。(1)受试者相关实验。在一个会话中，每个受试者的前 9 次试验用作源域，其余 6 次试验作为目标域。最终结果通过计算受试者间的平均准确率获得。(2)受试者无关实验。这里，我们采用了广泛使用的留一个受试者出去的策略。一个受试者一个会话的数据被用作目标域，而其余受试者一个会话的数据形成源域。例如，在 SEED 数据集中，每个受试者在一次会话中有 3394 个样本，导致源域大小为 14×3394 ，目标域大小为 1×3394 。我们重复此过程 15 次以获得每个受试者的结果，并报告所有受试者间的平均性能作为最终结果。这里，与先前研究[21]一样，SEED 数据集包含三个会话。第一个会话代表对刺激材料的初次观看，此时情绪反馈通常比后续会话更强。因此，我们的分析在主体相关和主体无关的情况下都集中在第一个会话。

C. 实施结果

所提出的 NSAL-DGAT 方法由两个主要组件构成：一个动态图注意力特征提取器（DGAT），称为 G，以及一个特征判别器，记为 G。DGAT 特征提取器 G 包含一个两层图卷积网络（GCN）和一个图注意力网络。特征判别器 G 实现为一个具有 64-64-2 层大小的多层感知器（MLP），并采用 ReLU 激活函数。此外，使用 RMSprop 优化器进行网络训练，学习率设置为 0.001，批量大小为 48。为减轻过拟合，网络中应用了 L1 正则化 ($1e-3$)。这些实验在 NVIDIA GeForce RTX 2080 GPU 上执行，使用 CUDA 10.0 和 PyTorch 进行高效的计算和深度学习操作。

D. 实验结果

在本节中，我们对两个数据集的两种场景进行了广泛的实验。实验结果如下所述。

1) SEED 的受主题影响的实验：为了评估为了验证所提出的 NSAL-DGAT 方法的有效性，我们采用受试者相关协议，并仔细选择两类具有代表性的基线方法进行大量实验。第一组包括传统方法，包括 SVM [39]、RF [40]、CCA [41]、GRSLR [25]、GSCCA [26]和深度信念网络（DBN）[36]。另一组包括深度学习模型，即 GCNN [14]、DANN [17]、DGCNN [29]、BIHDM [42]、RGNN [30]、BiDANN [32]、HDGCN [15]和 PR-PL [21]。

TABLE I

THE AVERAGE ACCURACIES (%) AND STANDARD DEVIATIONS (%) OF DIFFERENT METHODS ON THE SEED DATASET IN THE SUBJECT-DEPENDENT SCENARIO

Methods	A_{cc}	Methods	A_{cc}
<i>Traditional machine learning methods</i>			
SVM [39]	83.99±09.72	GRSLR [25]	87.39±08.64
RF [40]	78.46±11.77	GSCCA [26]	82.96±09.95
CCA [41]	77.63±13.21	DBN [36]	86.08±08.34
<i>Deep learning methods</i>			
GCNN [14]	87.40±09.20	DANN [17]	91.36±08.30
DGCNN [29]	90.40±08.49	BiHDM [42]	93.12±06.06
RGNN [30]	94.24±05.95	BiDANN [32]	92.38±07.04
HD-GCN [15]	96.40±04.54	PR-PL [21]	94.84±09.16
NSAL-DGAT	96.53±09.10		

TABLE II

THE AVERAGE ACCURACIES (%) AND STANDARD DEVIATIONS (%) OF DIFFERENT METHODS ON THE HIED DATASET IN THE SUBJECT-DEPENDENT SCENARIO

Methods	A_{cc}	Methods	A_{cc}
<i>Traditional machine learning methods</i>			
SVM*	51.16±20.45	RF*	52.10±15.80
TCA*	53.58±16.52	DBN*	51.64±18.90
<i>Deep learning methods</i>			
GCNN* [14]	55.42±18.61	DANN* [17]	60.15±16.58
DGCNN* [29]	64.25±19.25	PR-PL* [21]	72.62±15.95
NSAL-DGAT	79.32±19.78		

The results of these models we reproduced are indicated by “*”.

Table I summarizes the experimental results on the SEED dataset. The results reveal that the proposed NSAL-DGAT achieves the highest accuracy (96.53±09.10), outperforming the previously reported results by 0.13%. These findings suggest that deep learning methods significantly outperform traditional machine learning approaches, indicating that deep learning models can more effectively learn generalizable and discriminative features. Among GCN-based deep learning methods, the NSAL-DGAT exhibits significantly improved classification performance compared to other methods, including GCNN, DGCNN, RGNN, and HD-GCN. This result indicates that the dynamic graph attention in NSAL-DGAT alleviates the limitation of learning only the global “static” intrinsic relations between EEG channels, allowing it to learn generalized discriminative domain-invariant features. Furthermore, in contrast to DANN-based transfer learning methods, such as DANN, BiHDM, BiDANN, and PR-PL, the proposed NSAL-DGAT effectively avoids aligning only the global feature distribution while neglecting the local class-wise conditional distribution. As a result, it achieves significant improvements in classification accuracy.

2) *Subject-Dependent Experiment on HIED*: Table II shows experimental results on the HIED dataset. We compare these results with two categories of baseline methods: machine-learning methods: SVM [39], RF [40], TCA [43], DBN [36]; deep-learning methods: GCNN [14], DANN [17], DGCNN [29], and

TABLE III

THE AVERAGE ACCURACIES (%) AND STANDARD DEVIATIONS (%) OF DIFFERENT METHODS ON THE SEED DATASET IN THE SUBJECT-INDEPENDENT SCENARIO

Methods	A_{cc}	Methods	A_{cc}
<i>Traditional machine learning methods</i>			
SVM [39]	58.18±13.85	TCA [43]	63.64±14.88
SA [44]	69.00±10.89	GFK [45]	71.31±14.09
T-SVM [46]	72.53±14.00	TPT [47]	75.17±12.83
<i>Deep learning methods</i>			
DGCNN [29]	79.95±09.02	DANN [17]	81.65±09.92
RGNN [30]	85.30±06.72	BiHDM [42]	85.40±07.53
HD-GCN [15]	87.10±07.10	BiDANN [32]	83.28±09.60
PGCN [31]	84.59±08.68	UDDA [20]	88.10±06.54
R2G-STNN [28]	84.16±07.63	PR-PL [21]	93.06±05.12
NSAL-DGAT	95.69±05.24		

PR-PL [21]. The results show that the proposed NSAL-DGAT method outperforms all baselines, achieving an average accuracy of 79.32%. Notably, deep learning methods exhibit superior performance by capturing discriminative features for emotion classification compared to machine learning approaches. Furthermore, the NSAL-DGAT method significantly surpasses the GCN-based deep learning models (GCNN, DGCNN) and the DA-based deep learning models (DANN, PR-PL), suggesting that NSAL-DGAT effectively learns generalized and discriminative domain-invariant features for EEG-based emotion classification.

3) *Subject-Independent Experiment on SEED*: The subject-independent experiment is the most popular evaluation protocol in affective BCI systems, which effectively evaluates the stability and generalization of the model across different subject data. Table III summarizes the experimental results on the SEED dataset, comparing the performance of the proposed method with two categories of baseline methods. Machine learning methods include SVM [39], TCA [43], SA [44], GFK [45], transductive SVM (T-SVM) [46], and TPT [47], while deep learning approaches comprise DGCNN [14], DANN [17], RGNN [30], BiHDM [42], HD-GCN [15], BiDANN [32], PGCN [31], UDDA [20], R2G-STNN [28], and PR-PL [21]. The results show that the proposed NSAL-DGAT outperforms all baseline models, achieving the best performance with an accuracy of $95.69 \pm 05.24\%$, 2.63% higher than the best results reported in the literature. Compared with other GCN-based deep learning models (DGCNN, RGCNN, HD-GCN, PGCN), the NSAL-DGAT method demonstrates a more effective ability to learn generalizable domain-invariant features for emotion classification. Furthermore, when compared to existing DANN-based deep transfer learning networks (DANN, BiHDM, BiDANN), the proposed NSAL-DGAT successfully captures pairwise feature relationships between domains after domain adaptation, leading to superior classification performance.

4) *Subject-Independent Experiment on HIED*: Table IV summarizes the experimental results on the HIED dataset, comparing them against two baseline methods. Specifically, machine learning methods include SVM [39], TCA [43], SA [44],

TAA(%) SD(%)
DMSEED DS-D

表 I
S

Methods	A_{cc}	Methods	A_{cc}
<i>Traditional machine learning methods</i>			
SVM [39]	83.99±09.72	GRSLR [25]	87.39±08.64
RF [40]	78.46±11.77	GSCCA [26]	82.96±09.95
CCA [41]	77.63±13.21	DBN [36]	86.08±08.34
<i>Deep learning methods</i>			
GCNN [14]	87.40±09.20	DANN [17]	91.36±08.30
DGCNN [29]	90.40±08.49	BiHDM [42]	93.12±06.06
RGNN [30]	94.24±05.95	BiDANN [32]	92.38±07.04
HD-GCN [15]	96.40±04.54	PR-PL [21]	94.84±09.16
NSAL-DGAT		96.53±09.10	

TAA(%) SD(%)
DMHIED DS-D

表 II
S

Methods	A_{cc}	Methods	A_{cc}
<i>Traditional machine learning methods</i>			
SVM*	51.16±20.45	RF*	52.10±15.80
TCA*	53.58±16.52	DBN*	51.64±18.90
<i>Deep learning methods</i>			
GCNN* [14]	55.42±18.61	DANN* [17]	60.15±16.58
DGCNN* [29]	64.25±19.25	PR-PL* [21]	72.62±15.95
NSAL-DGAT		79.32±19.78	

The results of these models we reproduced are indicated by ‘*’.

表 I 总结了在 SEED 数据集上的实验结果。结果表明，所提出的 NSAL-DGAT 达到了最高准确率（96.53±09.10），比先前报道的结果高 0.13%。这些发现表明，深度学习方法显著优于传统机器学习方法，表明深度学习模型可以更有效地学习具有泛化性和判别性的特征。在基于 GCN 的深度学习方法中，NSAL-DGAT 与其他方法（包括 GCNN、DGCNN、RGNN 和 HD-GCN）相比，表现出显著提高的分类性能。这一结果表明，NSAL-DGAT 中的动态图注意力机制缓解了仅学习 EEG 通道之间全局“静态”内在关系的局限性，使其能够学习具有泛化性和判别性的领域不变特征。此外，与基于 DANN 的迁移学习方法（如 DANN、BiHDM、BiDANN 和 PR-PL）相比，所提出的 NSAL-DGAT 有效地避免了仅对全局特征分布进行对齐而忽略局部类别条件分布的情况。因此，它在分类准确率上取得了显著提高。

2) HIED 数据集的受主体影响实验：表 II 展示了在 HIED 数据集上的实验结果。我们将这些结果与两类基线方法进行比较：机器学习方法：SVM [39]、RF [40]、TCA [43]、DBN [36]；深度学习方法：GCNN [14]、DANN [17]、DGCNN [29]，以及

TAA(%) SD(%)
DMSEED DS-I

表 III
S

Methods	A_{cc}	Methods	A_{cc}
<i>Traditional machine learning methods</i>			
SVM [39]	58.18±13.85	TCA [43]	63.64±14.88
SA [44]	69.00±10.89	GFK [45]	71.31±14.09
T-SVM [46]	72.53±14.00	TPT [47]	75.17±12.83
<i>Deep learning methods</i>			
DGCNN [29]	79.95±09.02	DANN [17]	81.65±09.92
RGNN [30]	85.30±06.72	BiHDM [42]	85.40±07.53
HD-GCN [15]	87.10±07.10	BiDANN [32]	83.28±09.60
PGCN [31]	84.59±08.68	UDDA [20]	88.10±06.54
R2G-STNN [28]	84.16±07.63	PR-PL [21]	93.06±05.12
NSAL-DGAT		95.69±05.24	

PR-PL [21]。结果显示，所提出的 NSAL-DGAT 方法优于所有基线，平均准确率达到 79.32%。值得注意的是，与机器学习方法相比，深度学习方法通过捕获用于情绪分类的判别特征表现出更优越的性能。此外，NSAL-DGAT 方法显著超越了基于图卷积网络的深度学习模型（GCNN、DGCNN）和基于域适应的深度学习模型（DANN、PR-PL），表明 NSAL-DGAT 有效地学习了适用于脑电图情绪分类的泛化和判别领域不变特征。

3) 在 SEED 上的无受试者实验：
独立实验是情感脑机接口系统中最流行的评估协议，它有效地评估了模型在不同被试数据上的稳定性和泛化能力。表 III 总结了在 SEED 数据集上的实验结果，比较了所提方法与两类基线方法的性能。机器学习方法包括 SVM [39]、TCA [43]、SA [44]、GFK [45]、传递式 SVM (T-SVM) [46] 和 TPT [47]，而深度学习方法则包含 DGCNN [14]、DANN [17]、RGNN [30]、BiHDM [42]、HD-GCN [15]、BiDANN [32]、PGCN [31]、UDDA [20]、R2G-STNN [28] 和 PR-PL [21]。结果表明，所提的 NSAL-DGAT 方法优于所有基线模型，以 95.69 ± 05.24% 的准确率取得了最佳性能，比文献中报道的最佳结果高 2.63%。与其他基于 GCN 的深度学习方法（DGCNN、RGCNN、HD-GCN、PGCN）相比，NSAL-DGAT 方法在情感分类中学习泛化领域不变特征的能力更为有效。此外，与现有的基于 DANN 的深度迁移学习网络（DANN、BiHDM、BiDANN）相比，所提出的 NSAL-DGAT 在领域自适应后成功捕获了领域之间的成对特征关系，从而实现了更优的分类性能。

4) HIED 上的无主体实验：表 IV 总结了在 HIED 数据集上的实验结果，并与两种基线方法进行了比较。具体而言，机器学习方法包括 SVM [39]、TCA [43]、SA [44]，

TABLE IV
THE AVERAGE ACCURACIES (%) AND STANDARD DEVIATIONS (%) OF DIFFERENT METHODS ON THE HIED DATASET IN THE SUBJECT-INDEPENDENT SCENARIO

Methods	A_{cc}	Methods	A_{cc}
<i>Traditional machine learning methods</i>			
SVM [39]	45.92±10.33	TCA* [43]	58.24±15.32
SA* [44]	61.47±12.48	DBN* [36]	60.14±14.88
<i>Deep learning methods</i>			
DGCNN* [29]	49.95±17.33	DANN [17]	49.22±17.36
HCNN [48]	47.84±06.72	MADA [49]	50.69±18.45
EDANN [38]	67.94±14.02	PR-PL* [21]	70.88±14.10
NSAL-DGAT		75.82±13.35	

The results of these models we reproduced are indicated by '*'.

TABLE V
THE ABLATION STUDY OF THE PROPOSED NSAL-DGAT MODEL

<i>Ablation study about training strategy</i>	A_{cc}
w/o discriminator and neighborhood semantic aware learning	82.90±07.56
w/o neighborhood semantic aware learning	84.05±08.64
w/o dynamically graph attention	94.56±05.88
<i>About hyperparameter in NSAL</i>	A_{cc}
w/ fixed pseudo-labeling	88.42±07.55
w/o entropy weighting strategy	93.88±06.12
NSAL-DGAT	95.69±05.24

DBN [36], and deep learning methods contain DGCNN [12], DANN [17], HCNN [48], MADA [49], EDANN [38], and PR-PL [21]. The results reveal that our proposed NSAL-DGAT achieves the highest accuracy (75.82±13.35), which is 4.94% higher than the previously reported results. Notably, compared with the GCN-based deep learning method and the DANN-based deep transfer learning network, the proposed NSAL-DGAT can learn the dynamic brain functional networks and distribution differences under different emotions, helping to learn the discriminative domain-invariant features.

E. Ablation Study

We conducted extensive ablation experiments to deepen our understanding of the NSAL-DGAT approach. Specifically, we analyzed the dynamic graph attention, domain adaptation, and neighborhood semantic aware learning individually on the SEED dataset in the subject-independent scenario. All ablation experiments were executed using identical hyperparameters and have been presented in Table V.

As shown in Table V, it is evident that domain adversarial learning plays a crucial role in reducing individual differences by aligning the distributions across subjects, leading to a significant performance improvement. Excluding both the domain discriminator and neighborhood semantic aware learning causes the recognition accuracy to drop from 95.69% to 82.90%. This also suggests that the neighborhood semantic aware learning captures the pairwise relationships of feature representations between domains after domain adaptation, which helps enhance model performance, resulting in 11.61% improvement in emotion recognition (from 84.05% to 95.69%). Without considering

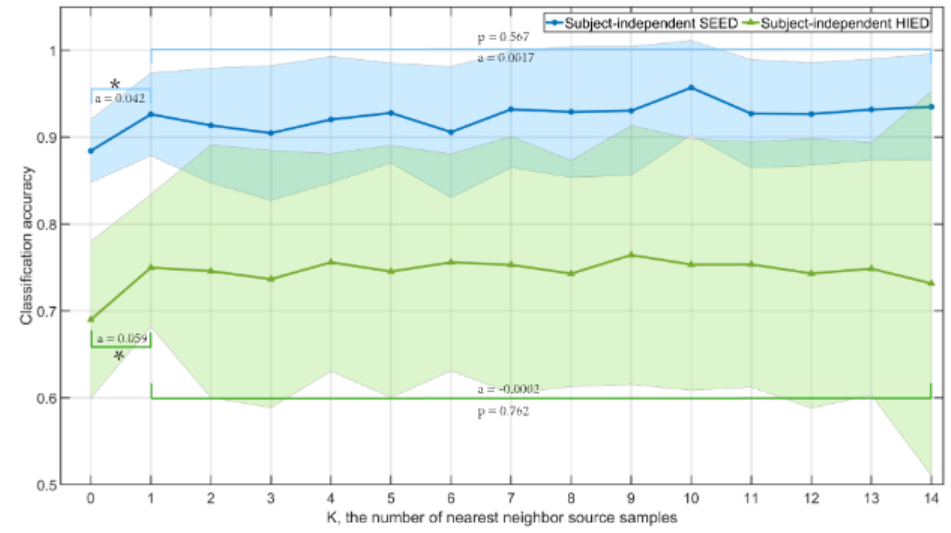


Fig. 4. Performance of different numbers of nearest neighbor source samples of the proposed NSAL-DGAT on the SEED and HIED datasets.

the graph attention convolutional layers, the model performance dropped by 1.13% (from 95.69% to 94.56%). This result shows that GAT can dynamically learn local brain function networks and contribute to learning discriminative features. In addition, to further elaborate on the effectiveness of NSAL, we examine the performance of different pseudo-labeling methods. When the pseudo-labeling method transitions from threshold-based to NSAL, model performance improves from 88.42% to 95.69%. Furthermore, we proposed an entropy weight strategy that further facilitates learning more accurate pseudo-labels for target domain data, leading to a 1.81% increase in emotion recognition rate (from 93.88% to 95.69%). These findings highlight the importance of these modules, and their combination can effectively improve EEG-based emotion recognition performance.

F. Number of Nearest Neighbor Source Samples

In NSAL-DGAT, we leveraged the nearest neighbor source domain samples to obtain the target semantic information. To explore the impact of the number of nearest neighbors on the performance of the proposed NSAL-DGAT method, we conducted more simulations on the SEED and HIED datasets under the subject-independent scenarios, and the experimental results are shown in Fig. 4.

From Fig. 4, we can observe that, in the subject-independent scenarios of the SEED and HIED datasets, when compared with the threshold-based pseudo-labeling ($K = 0$) method, the NSAL-based pseudo-labeling ($K = 1$) method significantly improves the emotion classification accuracy (SEED: $p < 0.05$, linear regression slope: $a = 0.042$; HIED: $p < 0.05$, linear regression slope: $a = 0.059$). This validates the superiority of NSAL in generating accurate pseudo-labels, directly enhancing subject-independent classification performance. As the number of the nearest neighbors increased from 1 to 14, we performed paired t-tests and linear regression slope analysis, and the statistical analysis showed no significant improvement in the emotion classification accuracy (SEED: $p = 0.567$, $a = 0.0017$; HIED: $p = 0.762$, $a = -0.0002$). However, a larger number of the nearest neighbors requires a higher computation load. Here, to balance robustness and computational complexity, we set K to 10.

表 IV
TAA(%) SD(%)
DMHIED DS-I
S

Methods	A_{cc}	Methods	A_{cc}
<i>Traditional machine learning methods</i>			
SVM [39]	45.92±10.33	TCA* [43]	58.24±15.32
SA* [44]	61.47±12.48	DBN* [36]	60.14±14.88
<i>Deep learning methods</i>			
DGCNN* [29]	49.95±17.33	DANN [17]	49.22±17.36
HCNN [48]	47.84±06.72	MADA [49]	50.69±18.45
EDANN [38]	67.94±14.02	PR-PL* [21]	70.88±14.10
NSAL-DGAT		75.82±13.35	

The results of these models we reproduced are indicated by ‘*’.

表 V
TASPNSAL-DGAT M

<i>Ablation study about training strategy</i>	A_{cc}
w/o discriminator and neighborhood semantic aware learning	82.90±07.56
w/o neighborhood semantic aware learning	84.05±08.64
w/o dynamically graph attention	94.56±05.88
<i>About hyperparameter in NSAL</i>	A_{cc}
w/ fixed pseudo-labeling	88.42±07.55
w/o entropy weighting strategy	93.88±06.12
NSAL-DGAT	95.69±05.24

DBN [36], 以及深度学习方法包含 DGCNN [12]、DANN [17]、HCNN [48]、MADA [49]、EDANN [38]和 PR-PL [21]。结果表明, 我们提出的 NSAL-DGAT 达到了最高准确率 (75.82±13.35), 比之前报道的结果高 4.94%。值得注意的是, 与基于 GCN 的深度学习方法和基于 DANN 的深度迁移学习网络相比, 提出的 NSAL-DGAT 能够学习不同情绪下的动态脑功能网络和分布差异, 有助于学习判别性域不变特征。

E. 消融研究

我们进行了广泛的消融实验, 以加深对 NSAL-DGAT 方法的理解。具体来说, 我们在受试者无关的场景下, 在 SEED 数据集上分别分析了动态图注意力、域适应和邻域语义感知学习。所有消融实验均使用相同的超参数执行, 结果已展示在表 V 中。如表 V 所示, 很明显领域对抗学习通过使不同受试者的分布对齐, 在减少个体差异方面起着关键作用, 从而显著提升了性能。排除领域判别器和邻域语义感知学习会导致识别准确率从 95.69%下降到 82.90%。这也表明邻域语义感知学习在领域适应后捕获了不同领域特征表示之间的成对关系, 这有助于提升模型性能, 使情感识别提高了 11.61% (从 84.05%提高到 95.69%)。不考虑

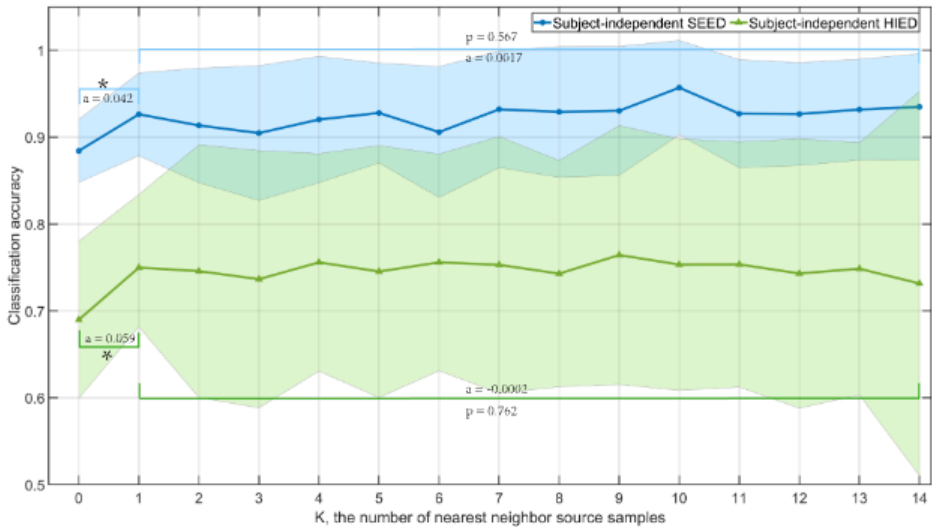


图 4. 所提出的 NSAL-DGAT 在不同数量的最近邻源样本下, 在 SEED 和 HIED 数据集上的性能表现。

图注意力卷积层, 模型性能下降了 1.13% (从 95.69%降至 94.56%)。这一结果表明 GAT 能够动态学习局部脑功能网络, 并有助于学习判别性特征。此外, 为了进一步阐述 NSAL 的有效性, 我们考察了不同伪标签方法的性能。当伪标签方法从基于阈值的过渡到 NSAL 时, 模型性能从 88.42%提升至 95.69%。此外, 我们提出了一种熵权策略, 进一步促进为目标域数据学习更准确的伪标签, 导致情绪识别率提高了 1.81% (从 93.88%升至 95.69%)。这些发现突出了这些模块的重要性, 它们的结合能够有效提高基于脑电图的情绪识别性能。

F. 最近邻源样本数量

在 NSAL-DGAT 中, 我们利用最近邻源域样本来获取目标语义信息。为了探究最近邻数量对所提出的 NSAL-DGAT 方法性能的影响, 我们在 SEED 和 HIED 数据集上进行了更多模拟, 在受试者无关的场景下, 实验结果如图 4 所示。从图 4 中我们可以观察到, 在 SEED 和 HIED 数据集的受试者无关场景下, 与基于阈值的伪标签方法 (K = 0) 相比, 基于 NSAL 的伪标签方法 (K = 1) 显著提高了情绪分类准确率 (SEED: $p < 0.05$, 线性回归斜率: $a = 0.042$; HIED: $p < 0.05$, 线性回归斜率: $a = 0.059$)。这验证了 NSAL 在生成准确伪标签方面的优越性, 直接提升了受试者无关分类性能。随着最近邻数量从 1 增加到 14, 我们进行了配对 t 检验和线性回归斜率分析, 统计分析显示情绪分类准确率没有显著提高 (SEED: $p = 0.567$, $a = 0.0017$; HIED: $p = 0.762$, $a = -0.002$)。然而, 更多的最近邻需要更高的计算负载。在此, 为平衡鲁棒性和计算复杂度, 我们将 K 设置为 10。

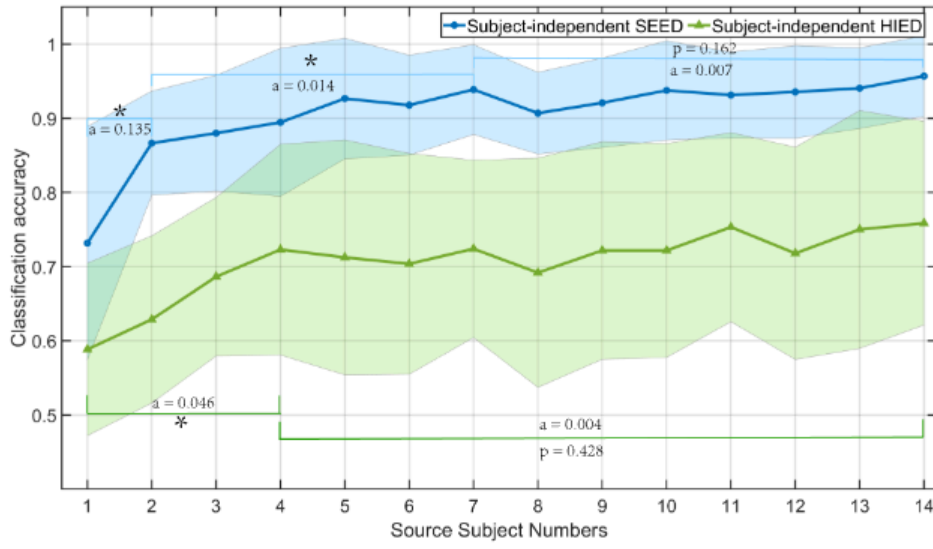


Fig. 5. Performance of different numbers of source subjects of the proposed NSAL-DGAT on the SEED and HIED datasets.

G. Number of Source Subjects

To evaluate the effect of the number of source subjects, more simulations have been conducted on the SEED and HIED datasets under the subject-independent scenarios. Specifically, we systematically calculated the average accuracy of 15 subjects under different numbers of source subjects, adding them in sequential order. Fig. 5 below illustrates the changes in the classification accuracy as the number of source subjects increases.

As shown in Fig. 5, we can observe that emotion classification accuracy exhibits a clear upward trend at the beginning with small source subject numbers and then almost a stable pattern with some fluctuations as the number of source-domain subjects further increases in subject-independent scenarios for both SEED and HIED datasets. Specifically, we performed the paired t-tests and calculated the linear regression slope “ a ” over intervals, which showed that the accuracy significantly improved when increasing the source subject number from 1 to 2 ($p < 0.05$, $a = 0.135$) and 2 to 7 ($p < 0.05$, $a = 0.014$) on the SEED dataset and 1 to 4 on the HIED dataset ($p < 0.05$, $a = 0.046$). This result suggests that diverse source domain data enhances the extraction of domain-invariant representations, resulting in significant improvements in cross-subject generalization. However, when the number of sources increases from 7 to 14 on the SEED dataset and 4 to 14 on the HIED dataset, our statistical analysis reveals no significant improvement in the classification accuracy (SEED: $p = 0.162$, $a = 0.007$; HIED: $p = 0.428$, $a = 0.004$). This result is reasonable, as the benefits of adding more source subjects diminish, which may be due to information saturation or potential issues such as negative transfer from weak correlation with the target domain [22], [35]. Overall, transferring from source subjects enables the NSAL-DGAT method to achieve significant performance improvements, rising from 73.18% to 95.69% (SEED) and from 59.47% to 75.82% (HIED), respectively.

H. Complexity and Computational Cost

To evaluate the efficiency and scalability of NSAL-DGAT, we compare its computational complexity and runtime performance against several GCN approaches. As shown in Table VI,

TABLE VI
COMPARISON OF MODEL COMPLEXITY BETWEEN NSAL-DGAT AND OTHER GCN MODELS ON THE SEED DATASET IN THE SUBJECT-INDEPENDENT SCENARIO

Methods	FLOPs	Params	Epoch(ms)	Acc $\pm$ Std(%)
DGCN [14]	821.7K	821.7K	6.147	79.95 $\pm$ 9.02
PGCN [31]	12426.7K	12286.1K	3.872	84.59 $\pm$ 8.68
NSAL-DGAT	1823.9K	294.7K	5.508	95.69 $\pm$ 5.24

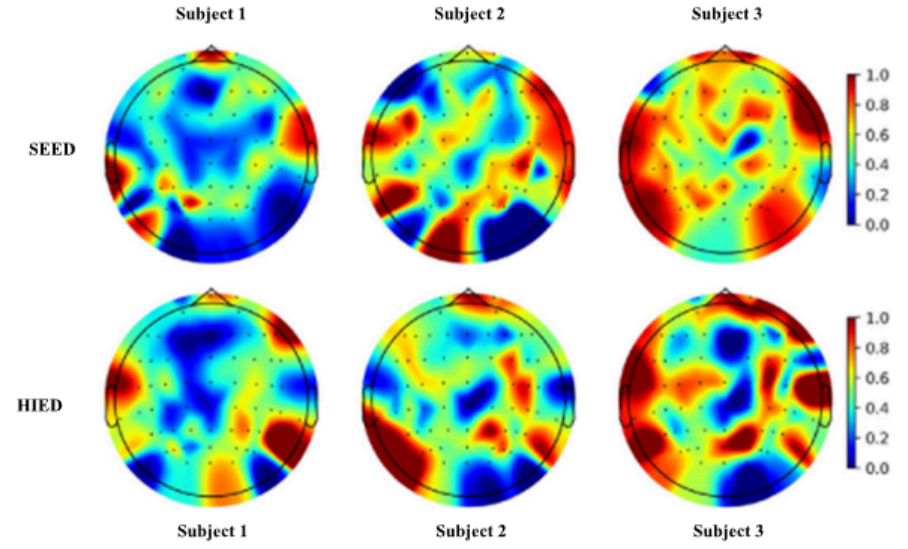


Fig. 6. Visualize the distribution of degree centrality of the global adjacency matrix learned by the proposed model on three subjects on the SEED and HIED datasets, respectively.

the proposed NSAL-DGAT exhibits significant parameter efficiency, with fewer trainable parameters than traditional GCN-based methods (DGCN, PGCN), facilitating easier real-world deployment. Besides, the average computation time of each data sample for emotion recognition is 0.0055 s with a standard deviation of 0.00042 s, confirming its feasibility in practical BCI applications [21].

V. DISCUSSION

In this section, we detail which global functional brain connectivity and local class-wise substructure benefit EEG-based emotion recognition. We then present the confusion matrix on three types of emotions to evaluate the model’s classification ability. Besides, we conduct the t-distributed stochastic neighbor embedding (T-SNE) visual analysis to validate the performance of NSAL-DGAT.

A. Distribution of Degree Centrality

The degree of centrality distribution visually represents the mapping of functional brain connections to brain topology. It characterizes the strength of the connection between each node and other nodes in the adjacency matrix. The calculation formula for degree centrality [15] is as follows:

$$C_i = \sum_{j=1}^n (A_{i,j}^G) + \sum_{k=1}^n (A_{i,k}^G) - 2A_{i,i}^G, \quad (i = 1, \dots, 62) \quad (15)$$

where n represents the number of EEG channels.

Fig. 6 illustrates the degree centrality distribution of the global adjacency matrix learned by the proposed model in subject-independent scenarios in the SEED and HIED datasets.

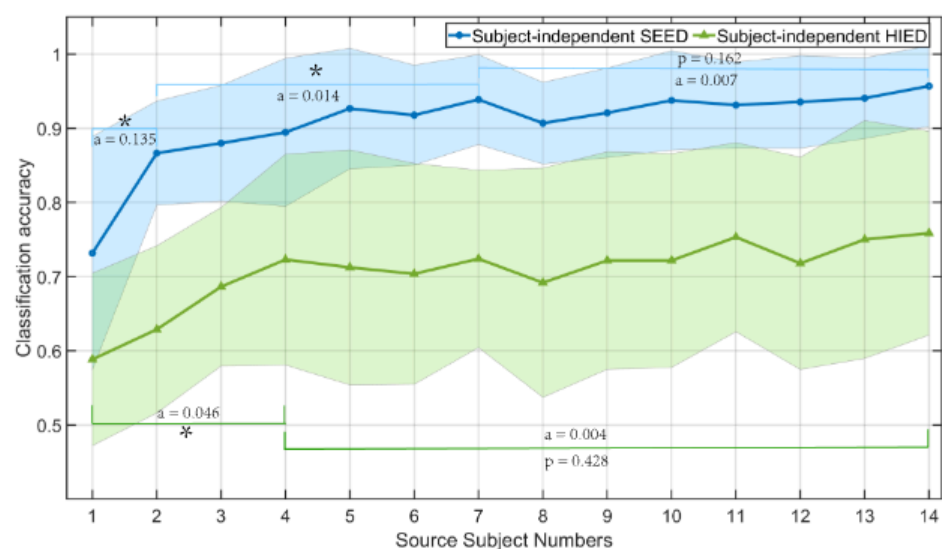


图 5. 所提出的 NSAL-DGAT 在不同源被试数量下在 SEED 和 HIED 数据集上的性能表现。

G. 源被试数量

为了评估源被试数量的影响，在受试者独立场景下，我们在 SEED 和 HIED 数据集上进行了更多模拟。具体而言，我们系统地计算了在不同源被试数量下 15 个被试的平均准确率，并按顺序依次增加。下图展示了随着源被试数量增加，分类准确率的变化情况。

如图 5 所示，我们可以观察到，在源主体数量较少时，情绪分类准确率呈现出明显的上升趋势，而当源域主体数量进一步增加时，在主体独立的场景下，SEED 和 HIED 数据集的准确率几乎保持稳定，并伴有波动。具体来说，我们进行了配对 t 检验，并计算了区间的线性回归斜率“a”，结果表明，在 SEED 数据集上，当源主体数量从 1 增加到 2 ($p < 0.05$, $a = 0.135$) 和从 2 增加到 7 ($p < 0.05$, $a = 0.014$) 时，准确率显著提高；在 HIED 数据集上，当源主体数量从 1 增加到 4 ($p < 0.05$, $a = 0.046$) 时，准确率显著提高。这一结果表明，多样化的源域数据增强了领域不变表示的提取，从而显著提高了跨主体的泛化能力。然而，当 SEED 数据集上的源主体数量从 7 增加到 14，以及 HIED 数据集上的源主体数量从 4 增加到 14 时，我们的统计分析显示分类准确率没有显著提高 (SEED: $p = 0.162$, $a = 0.007$; HIED: $p = 0.428$, $a = 0.004$)。这个结果很合理，因为增加更多源受试者的好处会逐渐减少，这可能是因为信息饱和或与目标域弱相关导致的负迁移等问题[22], [35]。总体而言，从源受试者迁移使 NSAL-DGAT 方法实现了显著的性能提升，分别从 73.18% 提升到 95.69% (SEED) 和从 59.47% 提升到 75.82% (HIED)。

H. 复杂度和计算成本

为了评估 NSAL-DGAT 的效率和可扩展性，我们将其计算复杂度和运行时性能与几种 GCN 方法进行了比较。如表 VI 所示，

表 VI
C MC B NSAL-DGAT O
GCN MSEED DS-I
S

Methods	FLOPs	Params	Epoch(ms)	Acc ± Std(%)
DGCN [14]	821.7K	821.7K	6.147	79.95±9.02
PGCN [31]	12426.7K	12286.1K	3.872	84.59±8.68
NSAL-DGAT	1823.9K	294.7K	5.508	95.69±5.24

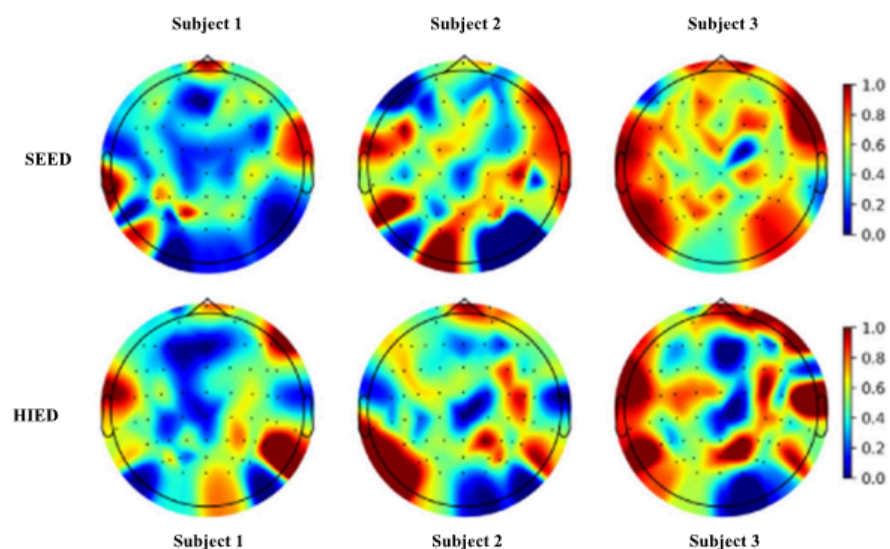


图 6. 在 SEED 和 HIED 数据集上，分别可视化所提出模型学习到的全局邻接矩阵的度中心性分布。

所提出的 NSAL-DGAT 表现出显著的参数效率，其可训练参数数量少于传统的基于 GCN 的方法 (DGCN、PGCN)，便于实际部署。此外，情绪识别每个数据样本的平均计算时间为 0.0055 秒，标准差为 0.00042 秒，证实其在实际 BCI 应用中的可行性[21]。

V. D 在本节中，我们详细阐述了哪些全局功能脑连接和局部类别子结构有利于基于脑电图的情绪识别。然后，我们展示了三种情绪类型的混淆矩阵以评估模型的分类能力。此外，我们进行了 t 分布随机邻域嵌入 (T-SNE) 的视觉分析以验证 NSAL-DGAT 的性能。

A. 度中心性分布

度中心性分布直观地表示功能脑连接到脑拓扑的映射。它表征了邻接矩阵中每个节点与其他节点之间连接的强度。度中心性的计算公式[15]如下：

$$C_i = \sum_{j=1}^n (A_{ij}) + \sum_{k=1}^n (A_{ki}) - 2A_{ii}, (i = 1, \dots, 62) \quad (15)$$

其中 n 表示 EEG 通道的数量。

图 6 展示了在 SEED 和 HIED 数据集中，独立于被试的情况下，所提出的模型学习到的全局邻接矩阵的度中心性分布。

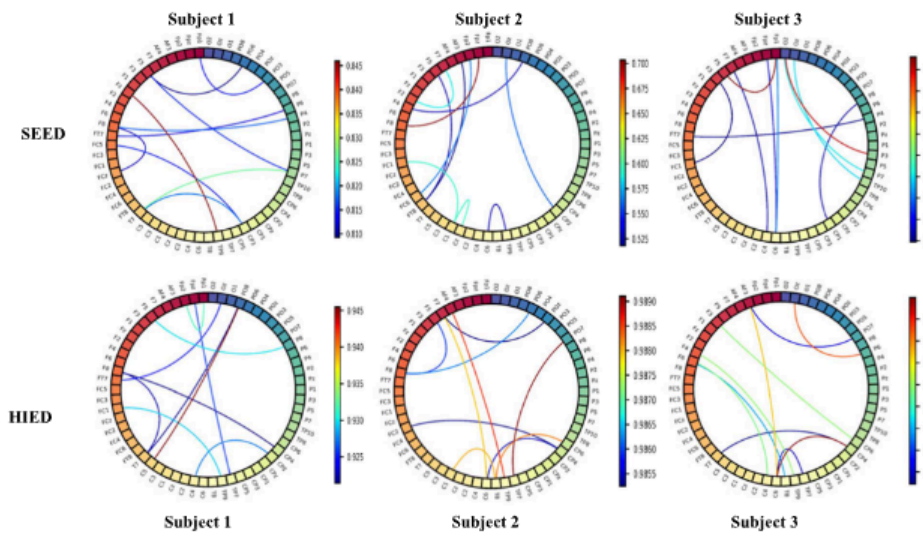


Fig. 7. Visualization of the top ten functional brain connections between EEG channels of the global adjacency matrix on the SEED and HIED datasets, respectively.

This analysis aims to investigate brain regions closely associated with EEG-based emotion classification. It is evident that there are significant differences in the distribution of electrodes with high connection strength between different subjects. However, it is noteworthy that despite the individual differences, the important electrode distributions of individuals with typical hearing and hearing-impaired individuals are mainly concentrated in the right prefrontal lobe and left temporal lobe. This observation suggests that these regions are strongly correlated with emotion cognition, and these findings are consistent with previous studies [22], [31], [37].

B. Global Functional Brain Connectivity

Fig. 7 provides a visual representation of the top ten connections between EEG channels derived from the global adjacency matrix learned by the proposed model in the subject-independent scenario using the SEED and HIED datasets. In each subfigure, the circle's periphery represents the location of the 62 electrodes, and the lines represent the connections between channels. The strength of the connection is indicated by the darkness of the red color. From Fig. 6, it is evident that the important electrode pairs differ for each subject. In the SEED dataset, the most important electrode pairs for the three subjects are (FZ, TP9), (FPZ, F8), and (FP1, F5). In the HIED dataset, the most important electrode pairs for the three subjects are (P08, C5), (PO7, TP7), and (C6, CP4). Notably, a larger proportion of electrodes from the prefrontal lobe and temporal lobe are observed in the SEED dataset, highlighting their crucial role in emotion recognition tasks. Conversely, in the HIED dataset, we observe that the occipital lobe and temporal lobe involve more electrodes, which may indicate that the temporal lobe and occipital lobe are closely related to emotion generation in hearing-impaired individuals [37], [50], [51], [52]. Moreover, it is worth noting that the electrode pairs come from different brain functional areas. This observation suggests that emotion generation depends on the coordination and cooperation between multiple brain regions. This result also demonstrates that GCN-based methods use graph theory to learn the stable functional connection of affective cognition, which helps us extract discriminative features and investigate cognitive functions.

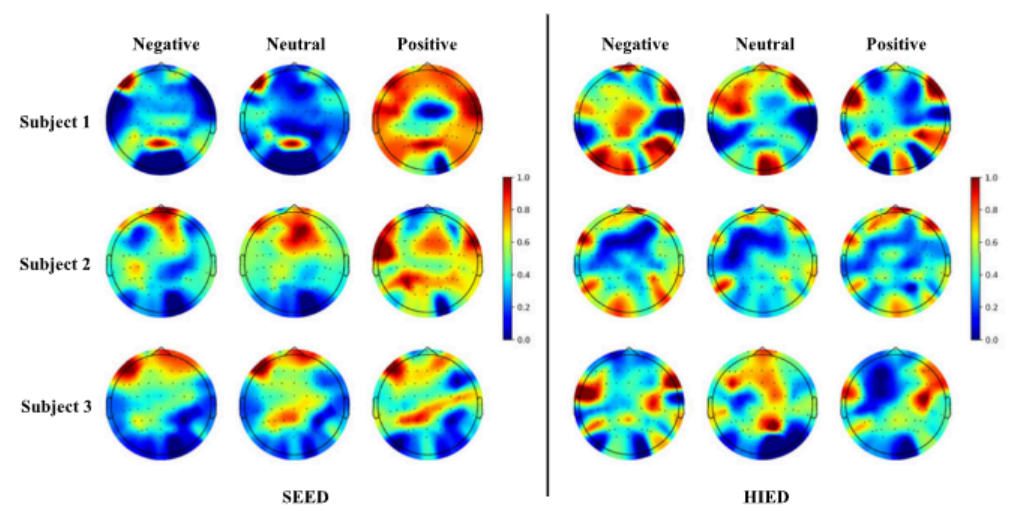


Fig. 8. Visualization of the important EEG channel weights obtained from DGAT for different emotional states on the SEED and HIED datasets, respectively.

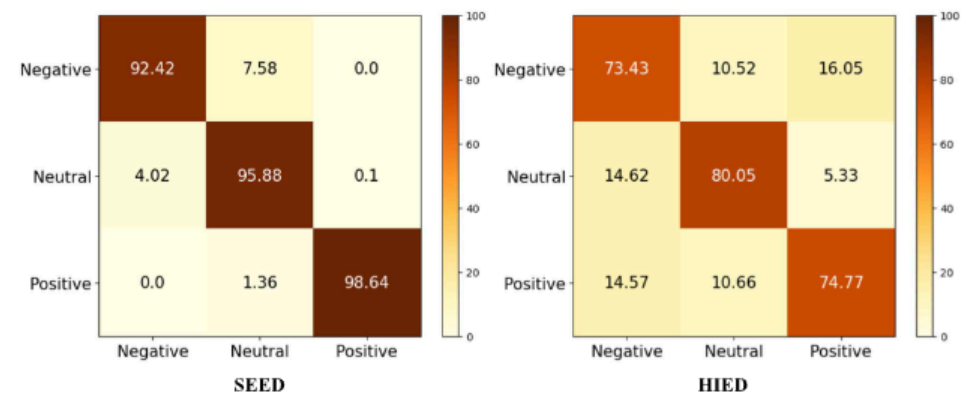


Fig. 9. Confusion matrices of NSAL-DGAT for SEED and HIED datasets in subject-independent experiments.

C. Class-Wise Local Brain Network

Fig. 8 shows the important EEG channels associated with different emotional states obtained by the DGAT in two datasets. Different subjects exhibit variations in important EEG channels. In the SEED dataset, the distribution of important channels for negative and neutral emotions appears similar, primarily concentrated in the prefrontal and temporal lobes. Similarly, in the HIED dataset, we observe a comparable distribution of important channels for negative and neutral emotions, primarily located in the prefrontal cortex. This evidence suggests that the frontal lobe cortex plays a crucial role in emotional responses for both individuals with typical hearing and those with hearing impairments. Moreover, this similarity in important pathways for negative and neutral emotions indicates potential common neural mechanisms underlying these emotional states.

D. Confusion Matrix

Fig. 9 displays the confusion matrices of the proposed NSAL-DGAT method in subject-independent experiments conducted on the SEED and HIED datasets, respectively. These matrices allow us to evaluate classification performance across three emotions. In the SEED dataset, NSAL-DGAT can distinguish positive emotions better than neutral and negative emotions. This finding suggests that positive emotions are more easily induced in each subject than the other two emotions. Notably, NSAL-DGAT tends to misclassify negative emotions as neutral emotions, indicating that neutral and negative emotions may share similar emotion perception patterns, or that significant

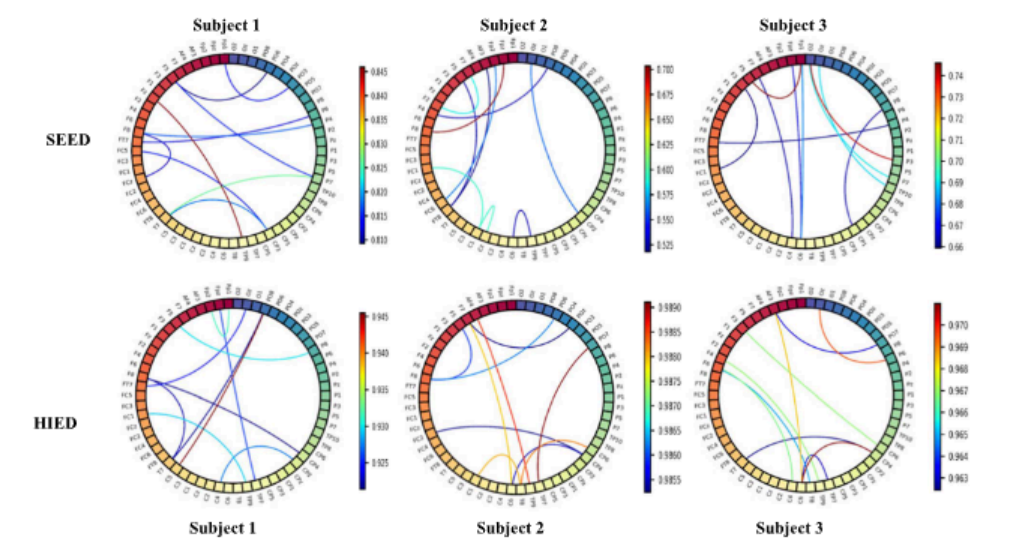


图 7. 在 SEED 和 HIED 数据集上，全局邻接矩阵中 EEG 通道之间前十位功能脑连接的可视化。

这项分析旨在研究与基于 EEG 的情绪分类密切相关的大脑区域。很明显，不同受试者之间高连接强度的电极分布存在显著差异。然而值得注意的是，尽管存在个体差异，典型听力者和听力障碍者的重要电极分布主要集中在右侧前额叶和左侧颞叶。这一观察结果表明这些区域与情绪认知密切相关，这些发现与先前研究 [22], [31], [37] 的结果一致。

B. 全局功能脑连接

图 7 展示了在独立于被试的情景下，使用 SEED 和 HIED 数据集，由所提出模型学习到的全局邻接矩阵推导出的 EEG 通道间前十位连接的视觉表示。在每个子图中，圆周代表 62 个电极的位置，线条代表通道间的连接。连接的强度由红色深浅表示。从图 6 可以看出，重要电极对因被试而异。在 SEED 数据集中，三个被试最重要的电极对分别是(FZ, TP9)、(FPZ, F8)和(FP1, F5)。在 HIED 数据集中，三个被试最重要的电极对分别是(P08, C5)、(PO7, TP7)和(C6, CP4)。值得注意的是，SEED 数据集中额叶和颞叶的电极比例较大，突显了它们在情绪识别任务中的关键作用。相反地，在 HIED 数据集中，我们观察到枕叶和颞叶涉及更多电极，这可能表明颞叶和枕叶与听力障碍者情绪生成密切相关 [37], [50], [51], [52]。此外，值得注意的是电极对来自不同的脑功能区域。这一观察表明情绪生成依赖于多个脑区之间的协调与合作。这一结果还表明基于 GCN 的方法利用图论学习情感认知的稳定功能连接，这有助于我们提取判别性特征并研究认知功能。

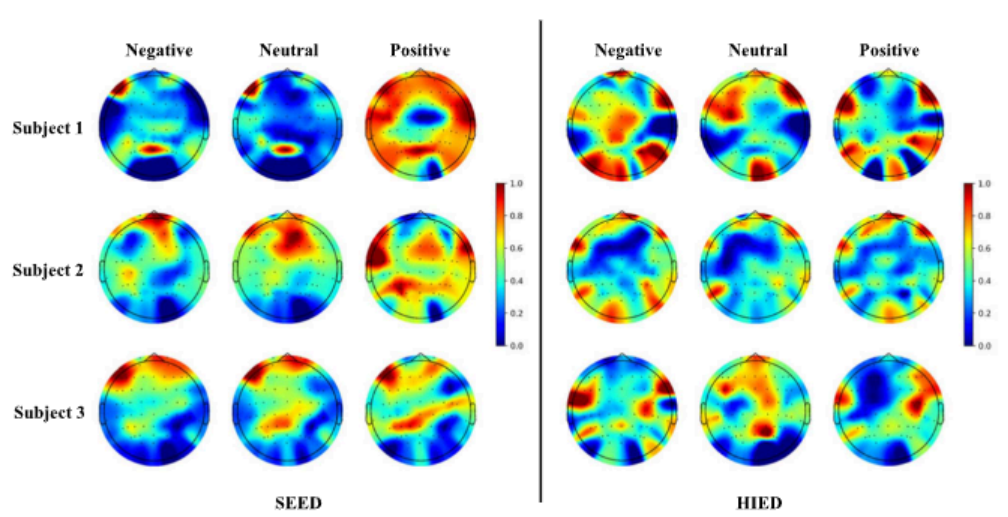


图 8. 分别在 SEED 和 HIED 数据集上，从 DGAT 获得的用于不同情绪状态的重要 EEG 通道权重可视化。

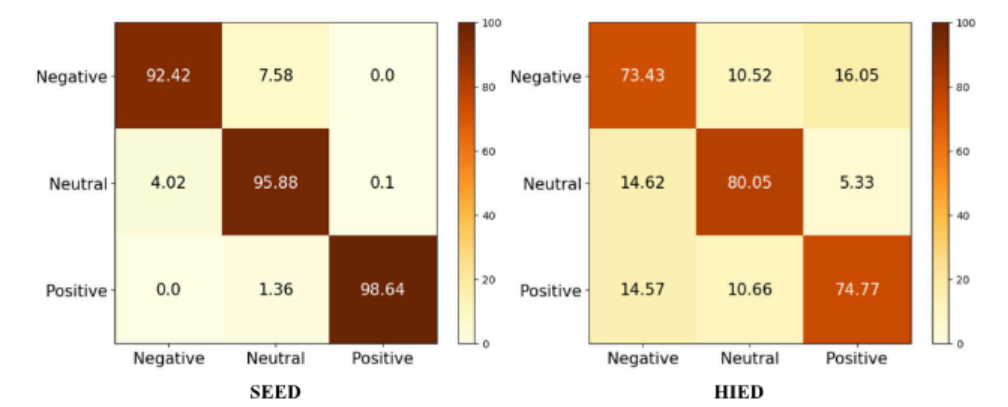


图 9. NSAL-DGAT 在受试者无关实验中针对 SEED 和 HIED 数据集的混淆矩阵。

C. 类别局部脑网络

图 8 展示了 DGAT 在两个数据集中获取的不同情绪状态下重要脑电图通道。不同受试者表现出重要脑电图通道的差异。在 SEED 数据集中，负面和中性情绪的重要通道分布相似，主要集中在前额叶和颞叶。同样，在 HIED 数据集中，我们观察到负面和中性情绪的重要通道分布相似，主要位于前额叶皮层。这一证据表明，额叶皮层在典型听力者和听力障碍者的情绪反应中均起着关键作用。此外，负面和中性情绪重要通路的一致性表明这些情绪状态可能存在潜在的共同神经机制。

D. 混淆矩阵

图 9 展示了在 SEED 和 HIED 数据集上分别进行的、针对独立受试者的实验中，所提出的 NSALDGAT 方法的混淆矩阵。这些矩阵使我们能够评估在三种情绪上的分类性能。在 SEED 数据集中，NSAL-DGAT 比中性情绪和负面情绪更能区分积极情绪。这一发现表明，积极情绪比其他两种情绪更容易在每位受试者中引发。值得注意的是，NSAL-DGAT 倾向于将负面情绪误分类为中性情绪，这表明中性情绪和负面情绪可能具有相似的感知模式，或者存在显著

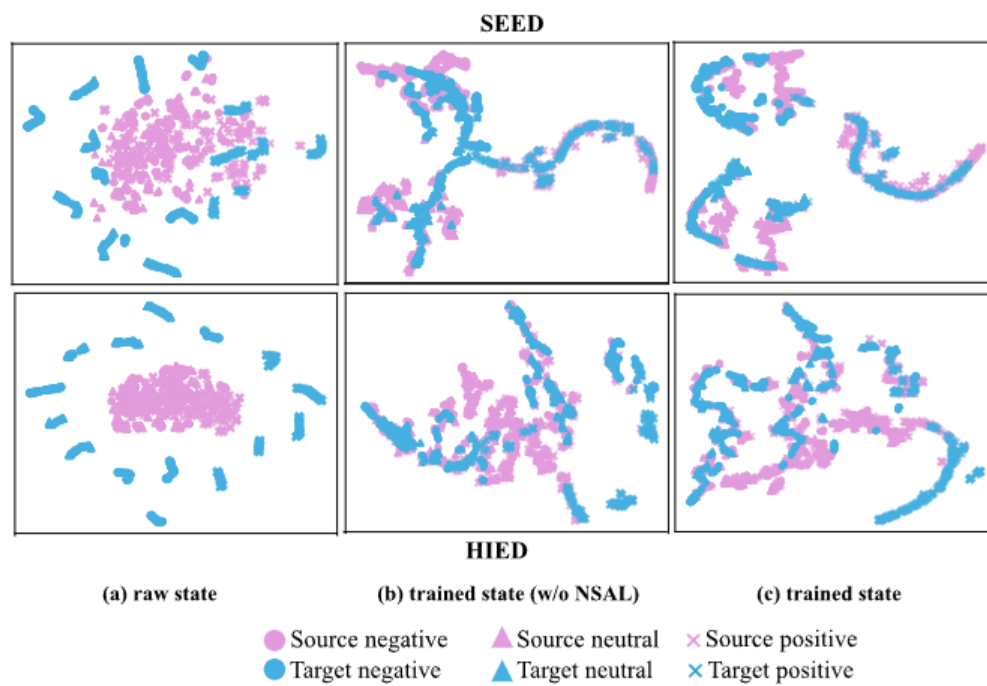


Fig. 10. Visualization of representative subjects for initial embeddings and trained embeddings without and with the NSAL strategy for subject-independent experiments on SEED and HIED datasets.

individual variability in negative emotions across different individuals makes it difficult to distinguish. On the HIED dataset, neutral emotions are more readily distinguished than negative and positive emotions. This observation may imply that hearing-impaired individuals face challenges in inducing positive and negative emotions due to their impairment. Besides, negative emotion samples are often misclassified as neutral or positive emotions by NSAL-DGAT. This suggests that the induction of negative emotions may be more dependent on musical stimuli. Overall, the confusion matrices in both datasets provide insights into the classification performance of NSAL-DGAT for different emotions. The results highlight differences in inducing and classifying specific emotional states, shedding light on the underlying patterns associated with emotion recognition in different populations.

E. T-SNE Visualization

Fig. 10 shows the visualization of the raw data distribution, trained model without using NSAL, and trained model using NSAL on the SEED and HIED datasets, offering a more intuitive insight into the classification performance and efficacy of the proposed NSAL-DGAT method. The T-SNE technology is used to showcase the raw and post-training feature distributions. Here, 1000 samples are randomly selected from both the source and target domains for feature visualization.

From the experimental results, it can be seen that there is a significant feature distribution discrepancy between the raw source and target domain data. After implementing the proposed NSAL-DGAT method without and with the NSAL strategy, the domain distributions are both aligned. Furthermore, compared to without the NSAL strategy, the proposed NSAL-DGAT method makes the aligned source and target domain data more compactly clustered under the same emotion, and there are clear boundaries between different emotion categories, which indicates that each emotion has unique patterns and characteristics. This observation suggests that the proposed NSAL-DGAT method effectively enhances the separation between different emotion categories

while preserving the similarity within each category. This alignment of distributions helps distinguish emotions more clearly and facilitates model generalization across domains, thereby improving the performance of emotion classification.

Overall, the visualization results demonstrate the effectiveness of NSAL-DGAT in aligning data distributions and improving classification performance in the SEED and HIED datasets.

VI. CONCLUSION

The study proposes a neighborhood semantic-aware learning-based dynamic graph attention convolution (NSAL-DGAT) algorithm for learning discriminative domain-invariant representations for classification. The proposed NSAL-DGAT algorithm is evaluated on two EEG emotion databases (SEED and HIED datasets), using different evaluation protocols (subject-dependent and subject-independent scenarios). The extensive experimental results demonstrate that the proposed NSAL-DGAT achieves promising results on both evaluation protocols. Furthermore, the paper conducts an interpretability analysis, providing insights into the affective cognition patterns between healthy and hearing-impaired individuals. In the future, we will further analyze the differences in affective cognition between people with typical hearing and people with hearing loss.

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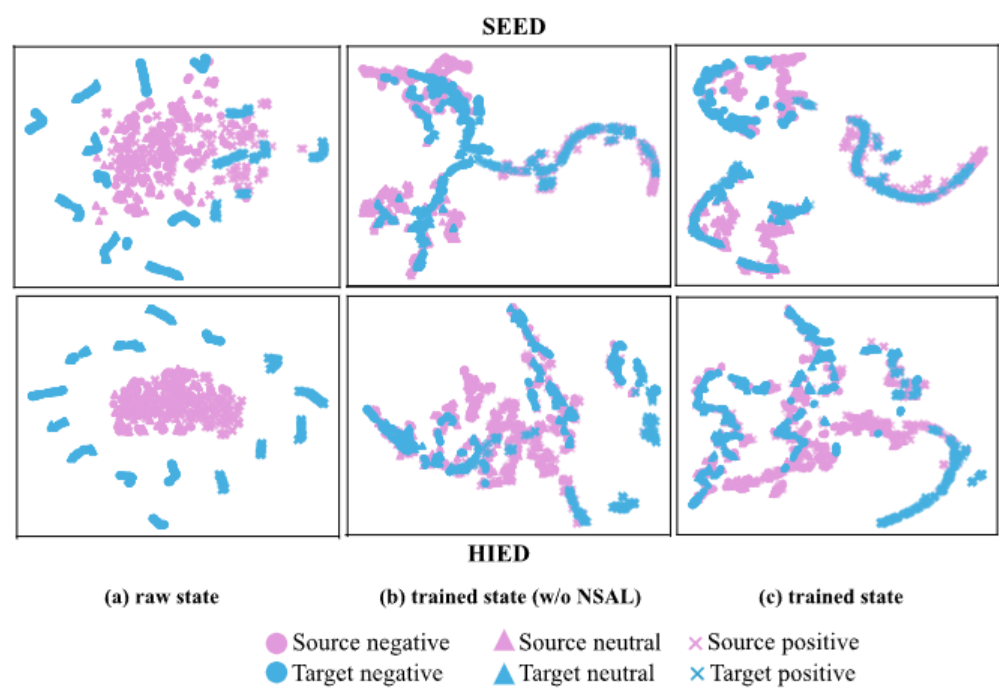


图 10. 在 SEED 和 HIED 数据集上进行的与受试者无关的实验中，初始嵌入和训练嵌入的可视化，展示了代表性受试者使用和不使用 NSAL 策略的情况。

不同个体之间负面情绪的个体差异使其难以区分。在 HIED 数据集上，中性情绪比负面和正面情绪更容易区分。这一观察可能表明，由于听力障碍，听力障碍个体在诱导正面和负面情绪方面面临挑战。此外，NSAL-DGAT 常常将负面情绪样本误分类为中性或正面情绪。这表明负面情绪的诱导可能更多地依赖于音乐刺激。总体而言，两个数据集的混淆矩阵为 NSAL-DGAT 的不同情绪分类性能提供了见解。结果突出了诱导和分类特定情绪状态之间的差异，揭示了与不同人群情绪识别相关的潜在模式。

E. T-SNE 可视化

图 10 展示了在 SEED 和 HIED 数据集上原始数据分布、未使用 NSAL 训练的模型以及使用 NSAL 训练的模型的可视化结果，为提出的 NSAL-DGAT 方法的分类性能和有效性提供了更直观的洞察。T-SNE 技术用于展示原始和训练后特征分布。此处，从源域和目标域中随机选择 1000 个样本进行特征可视化。从实验结果可以看出，原始源域和目标域数据之间存在显著的特征分布差异。在实施所提出的 NSAL-DGAT 方法（不使用和不使用 NSAL 策略）后，域分布均得到对齐。此外，与不使用 NSAL 策略相比，所提出的 NSAL-DGAT 方法使对齐的源域和目标域数据在相同情绪下更加紧凑地聚集，并且不同情绪类别之间存在清晰的边界，这表明每种情绪都有独特的模式和特征

这一观察表明，所提出的 NSAL-DGAT 方法能够有效增强不同情绪类别之间的分离，同时保持每个类别内部的相似性。这种分布的匹配有助于更清晰地区分情绪，并促进跨领域的模型泛化，从而提高情绪分类的性能。

总体而言，可视化结果展示了 NSAL-DGAT 在匹配数据分布和提高 SEED 和 HIED 数据集分类性能方面的有效性。

VI. C 该研究提出了一种基于邻域语义感知学习的动态图注意力卷积（NSALDGAT）算法，用于学习具有判别性和领域不变性的表征进行分类。所提出的 NSAL-DGAT 算法在两个脑电图情绪数据库（SEED 和 HIED 数据集）上进行了评估，采用了不同的评估协议（受试者相关和受试者无关场景）。广泛的实验结果表明，所提出的 NSAL-DGAT 在两种评估协议下均取得了良好的结果。此外，该论文进行了可解释性分析，为健康人群和听力障碍人群之间的情感认知模式提供了见解。未来，我们将进一步分析正常听力人群和听力损失人群之间的情感认知差异。

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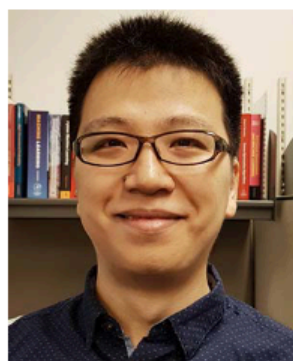
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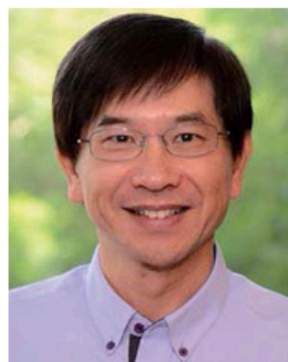


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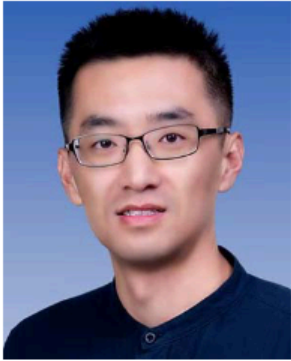
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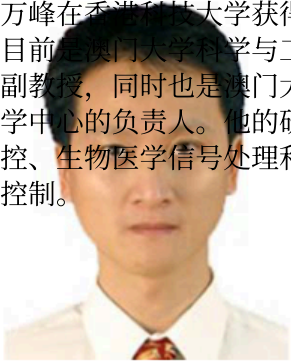
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